

BFS Traversal

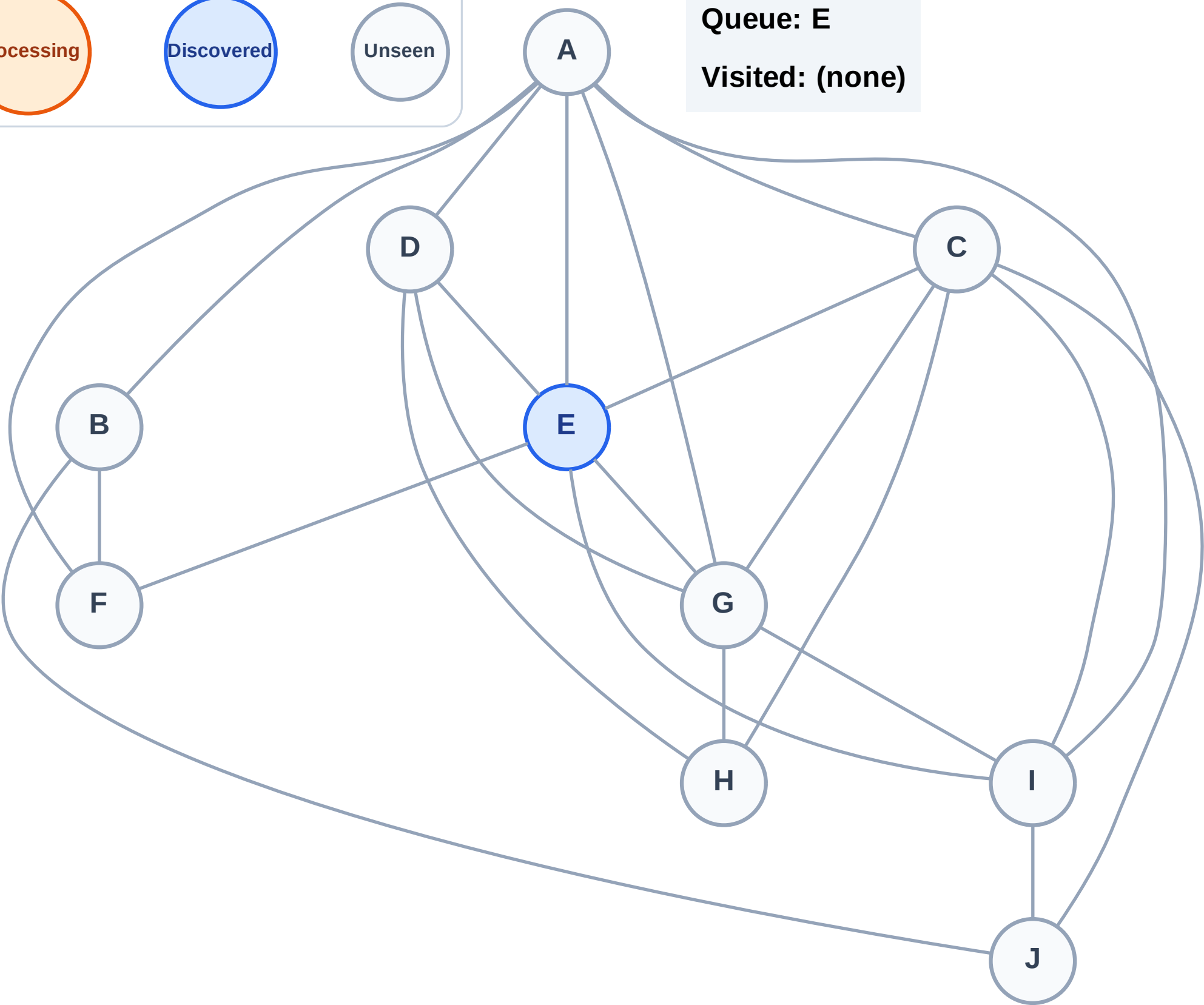
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

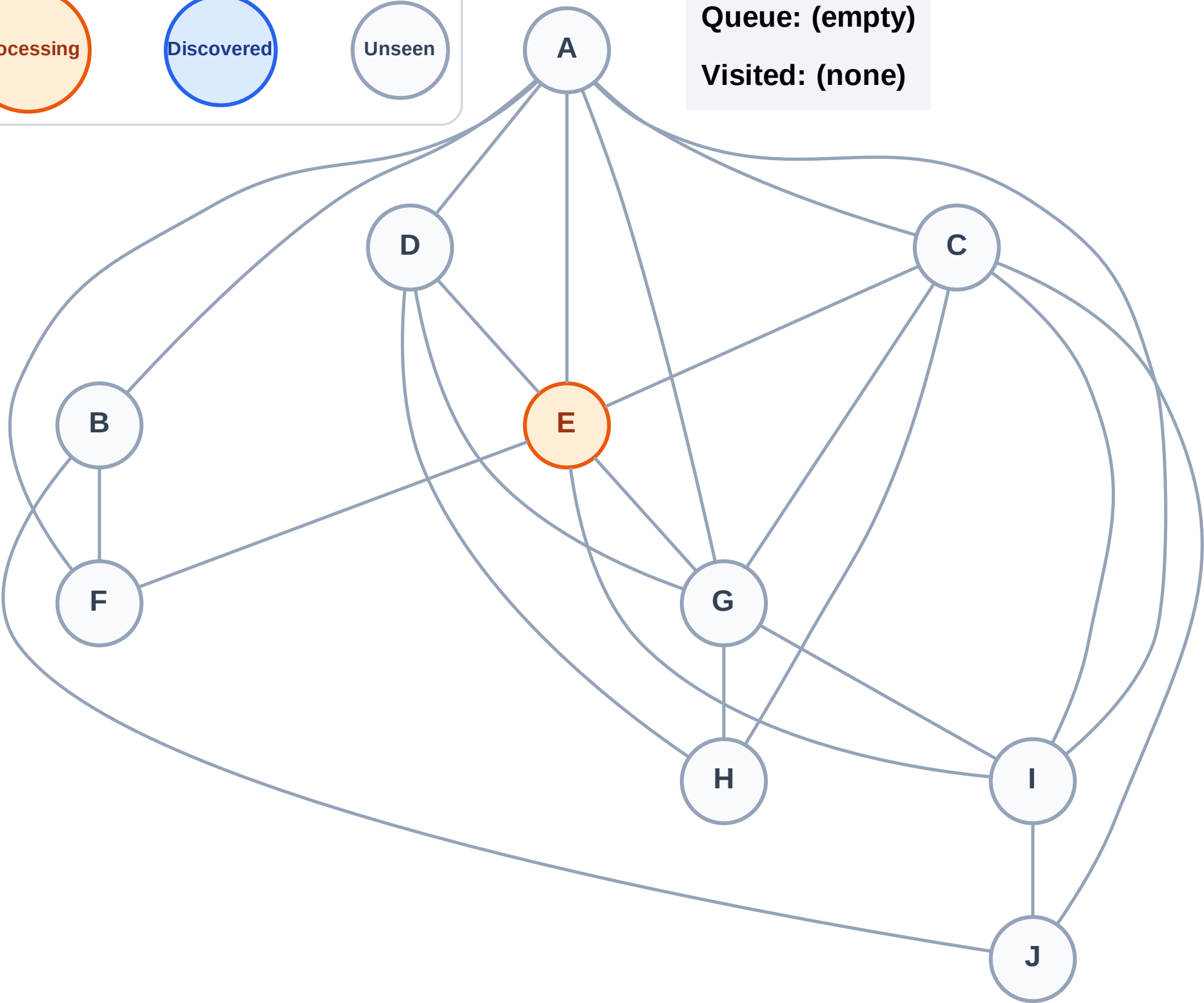
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



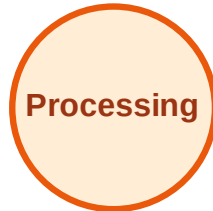
BFS Traversal

Step 3: Discover A, C, D, F, G, I from E

Legend



Complete



Processing



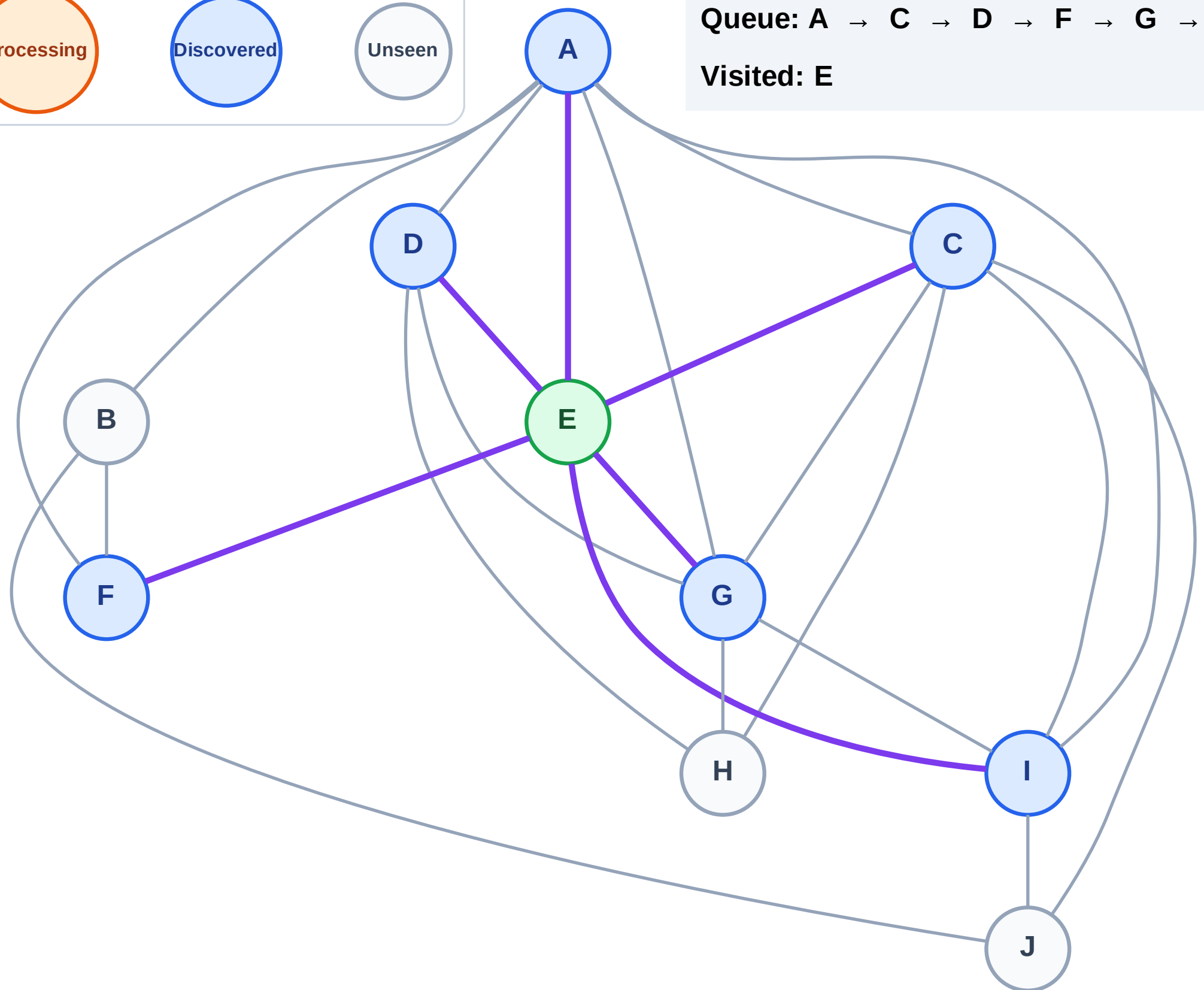
Discovered



Unseen

Queue: A → C → D → F → G → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

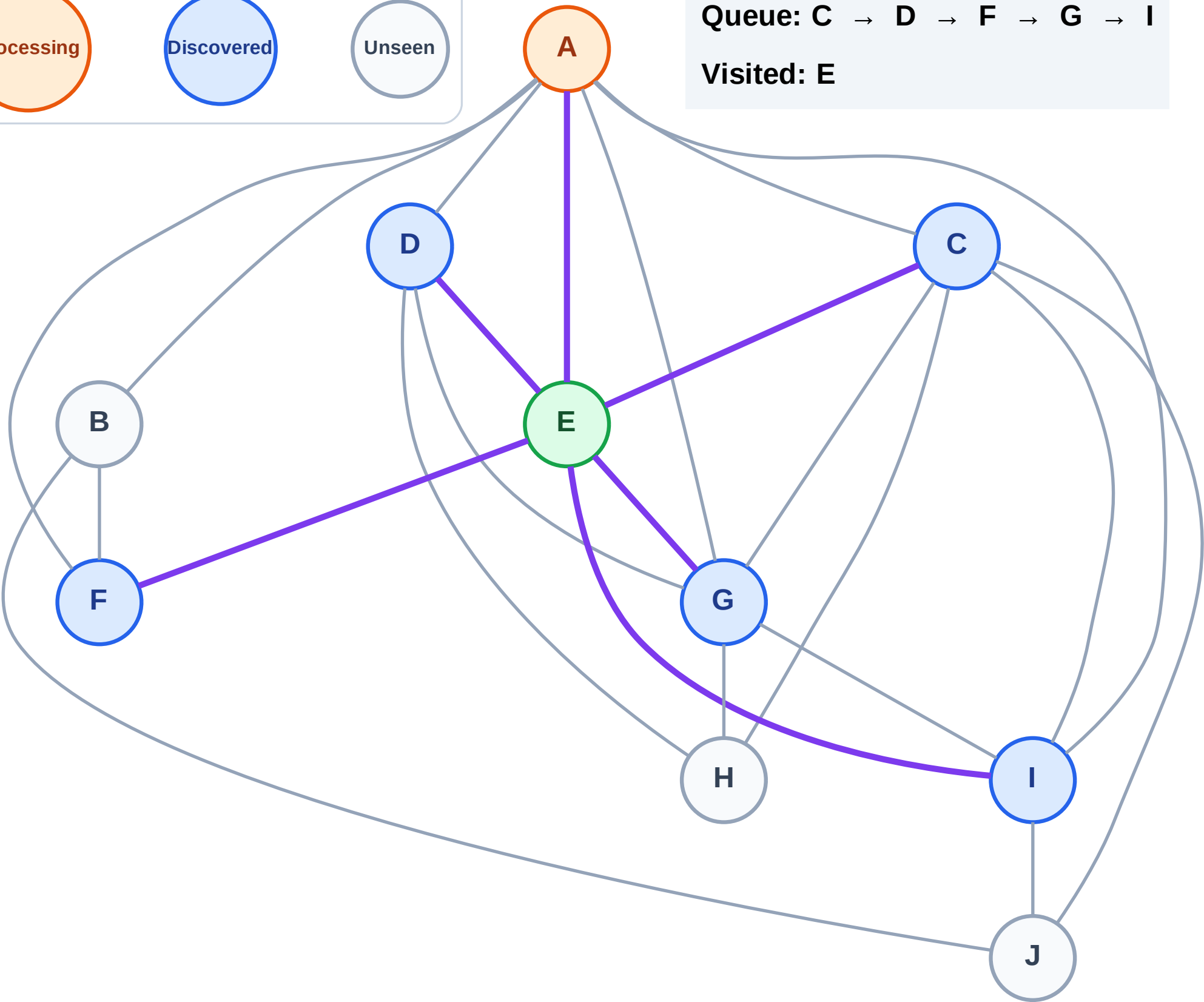
Processing

Discovered

Unseen

Queue: C → D → F → G → I

Visited: E



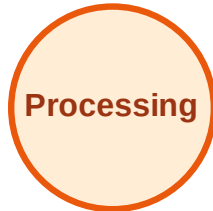
BFS Traversal

Step 5: Discover B from A

Legend



Complete



Processing



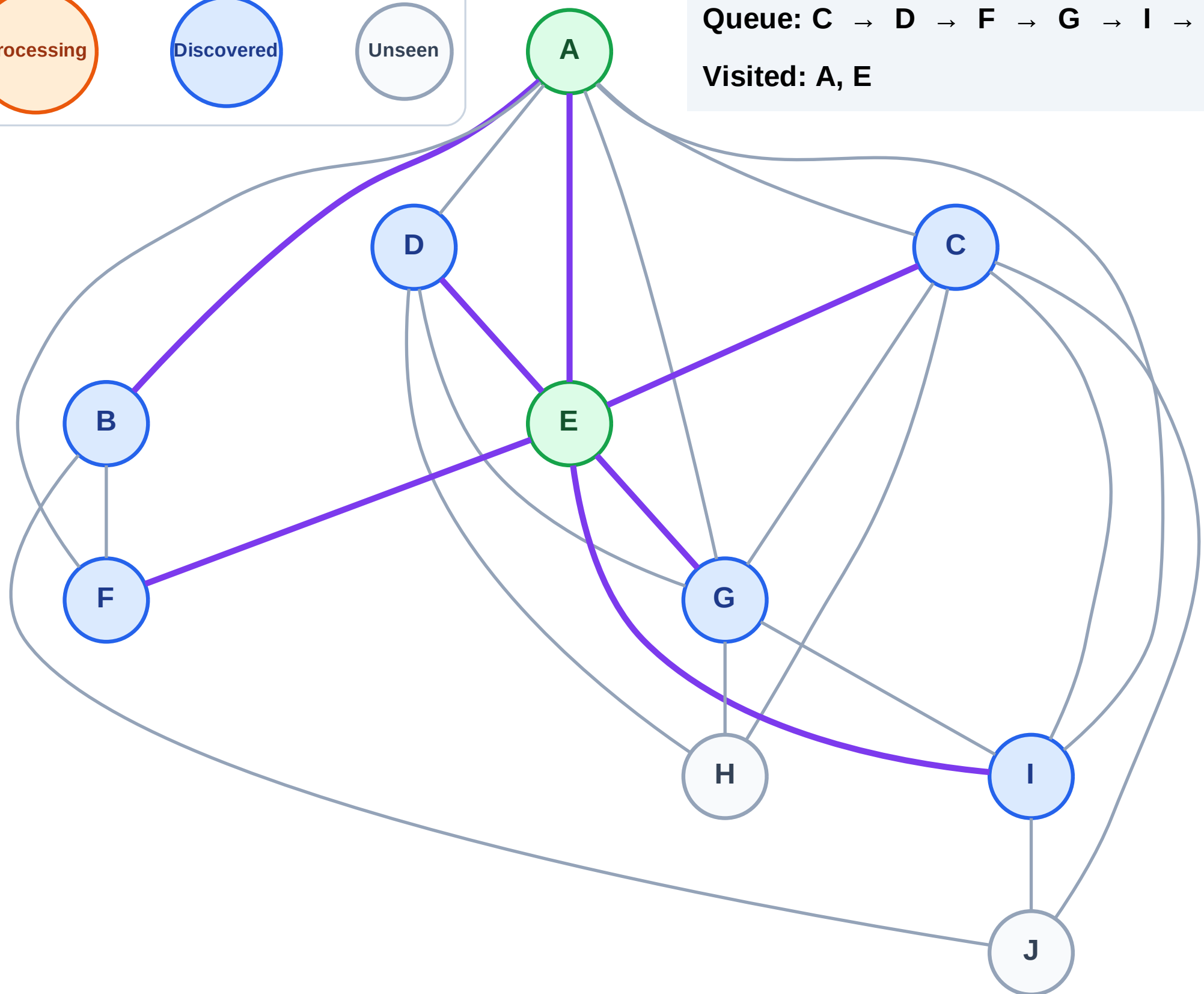
Discovered



Unseen

Queue: C → D → F → G → I → B

Visited: A, E



BFS Traversal

Step 6: Process node C

Legend

Complete

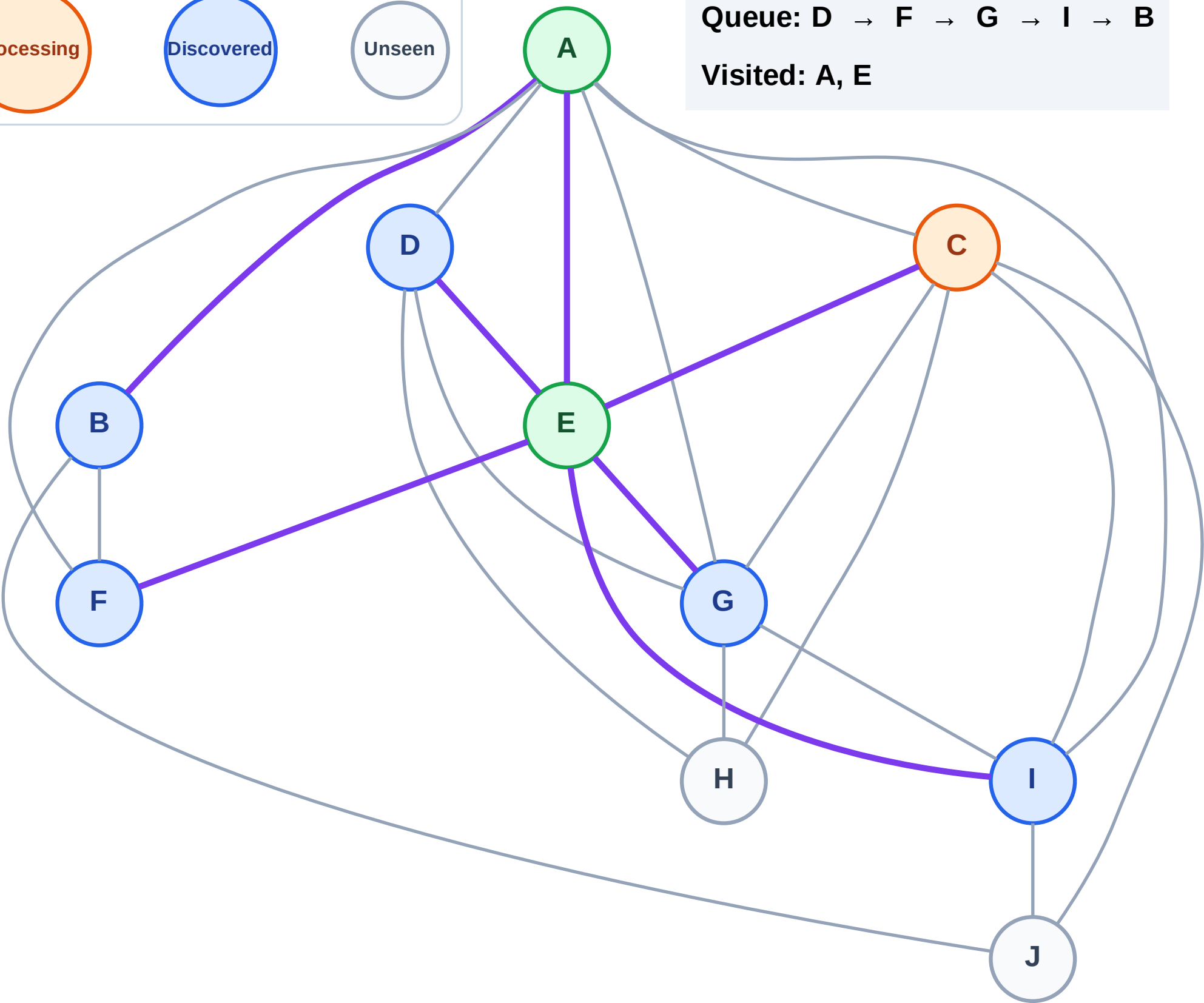
Processing

Discovered

Unseen

Queue: D → F → G → I → B

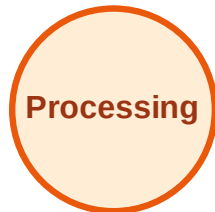
Visited: A, E



BFS Traversal

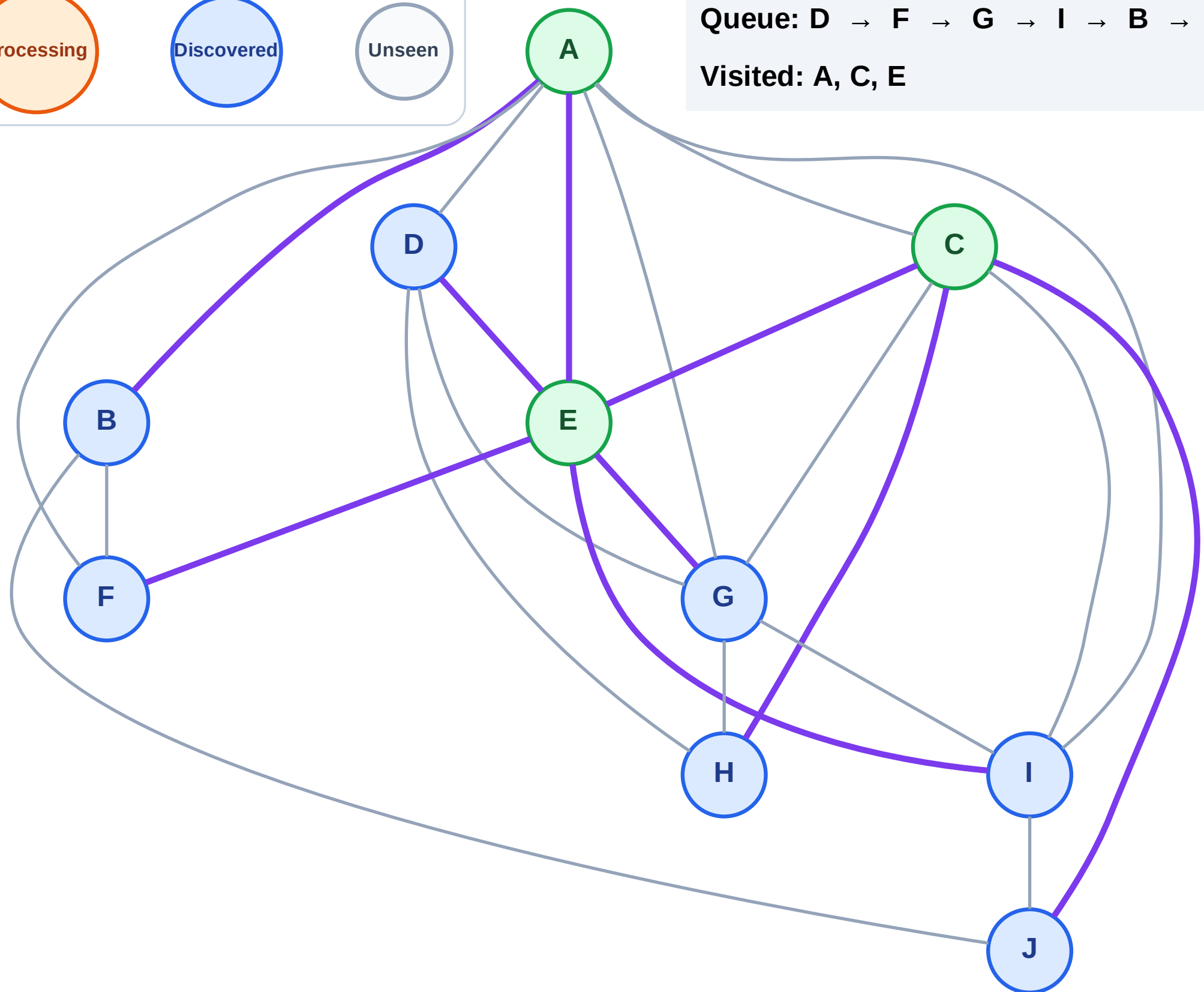
Step 7: Discover H, J from C

Legend



Queue: D → F → G → I → B → H → J

Visited: A, C, E



BFS Traversal

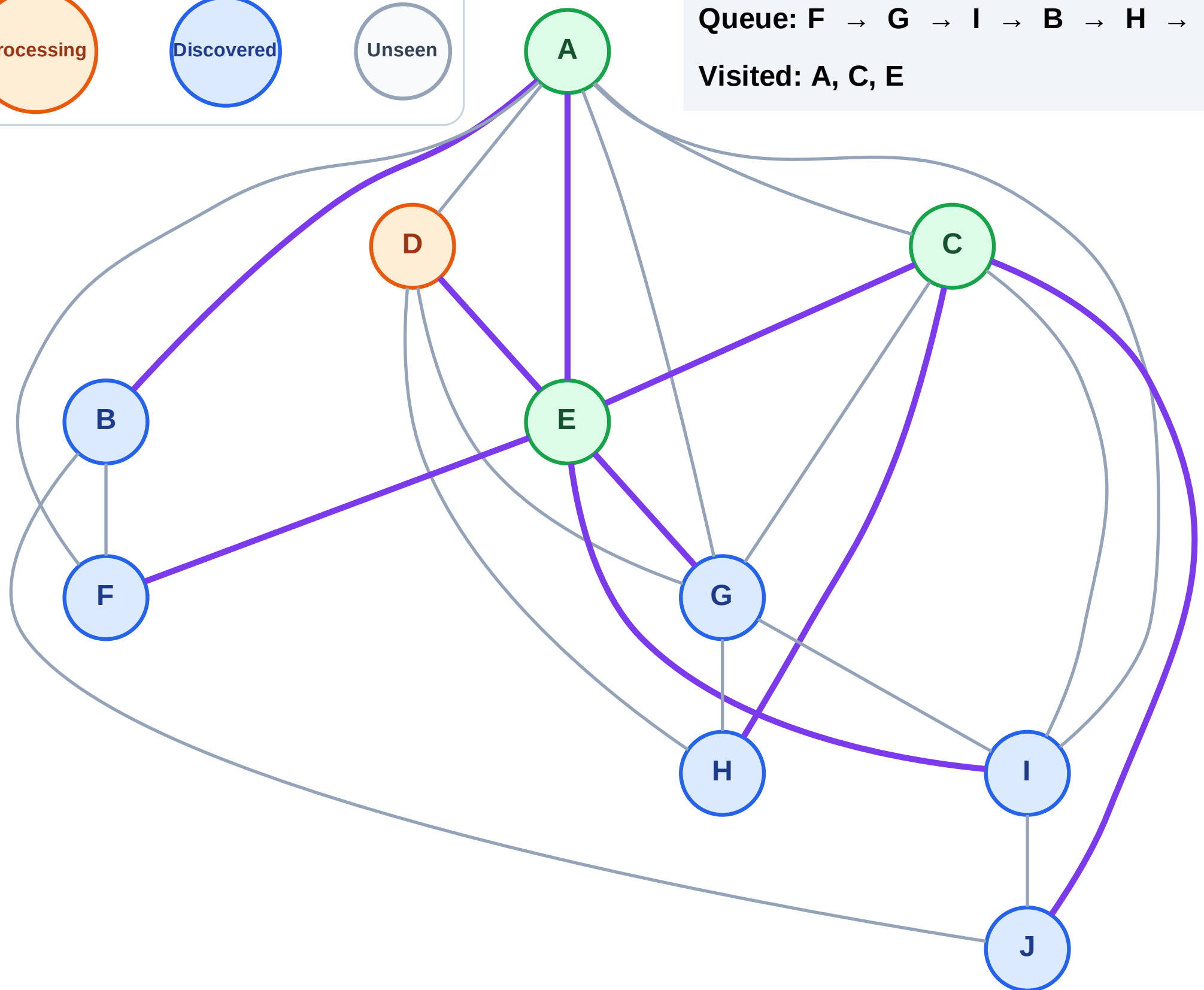
Step 8: Process node D

Legend



Queue: F → G → I → B → H → J

Visited: A, C, E



BFS Traversal

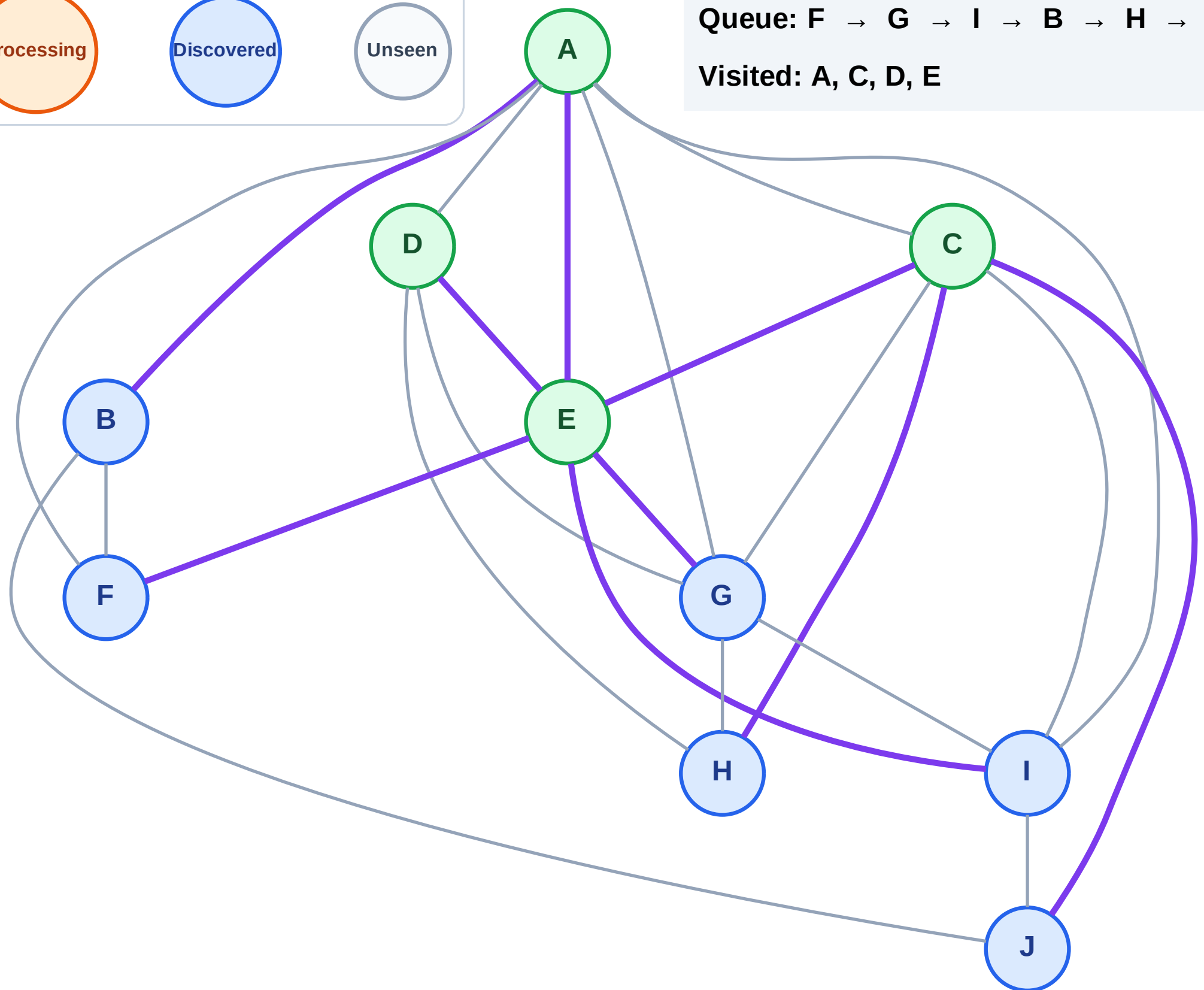
Step 9: No new neighbors from D

Legend



Queue: F → G → I → B → H → J

Visited: A, C, D, E



BFS Traversal

Step 10: Process node F

Legend

Complete

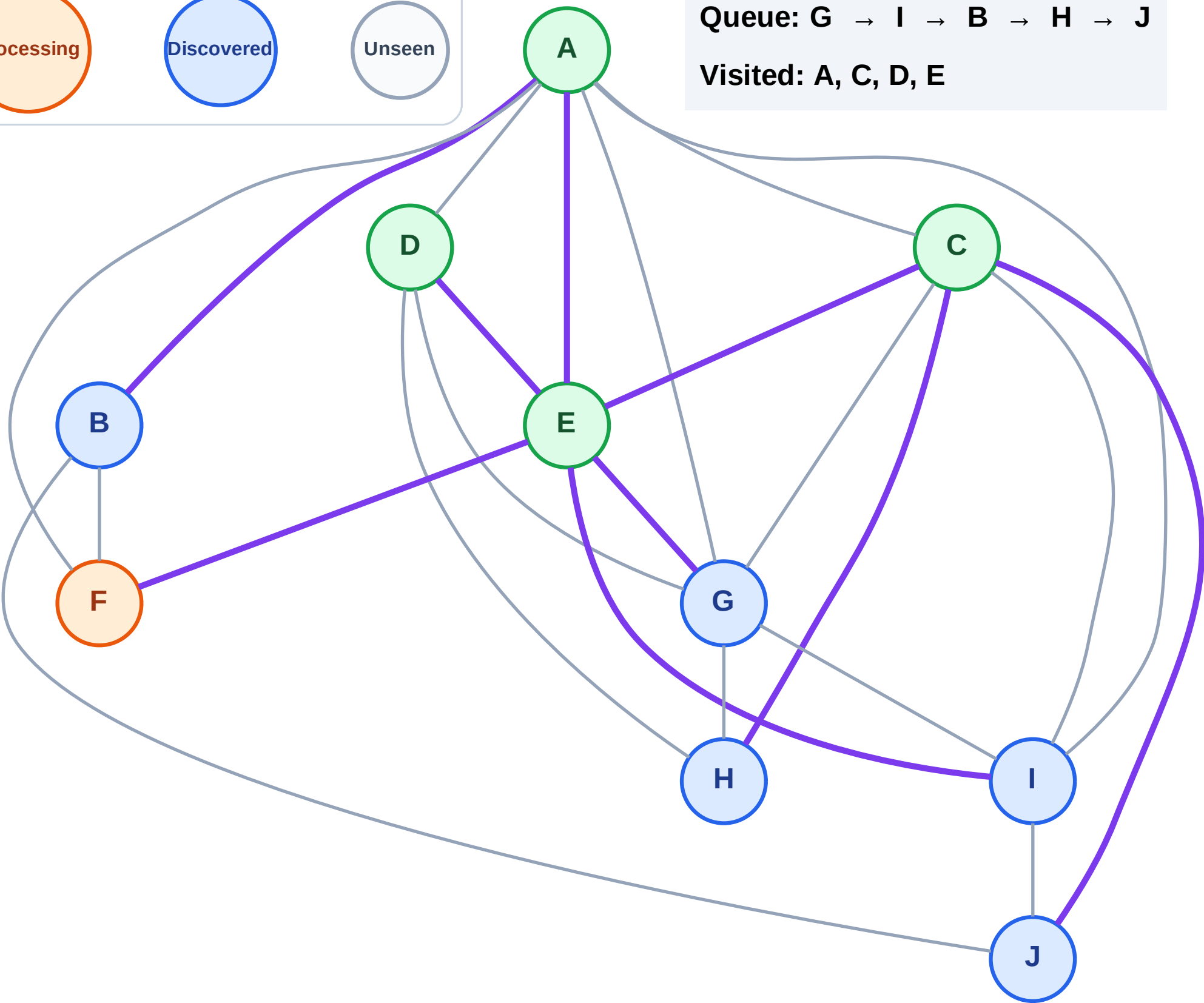
Processing

Discovered

Unseen

Queue: G → I → B → H → J

Visited: A, C, D, E



BFS Traversal

Step 11: No new neighbors from F

Legend

Complete

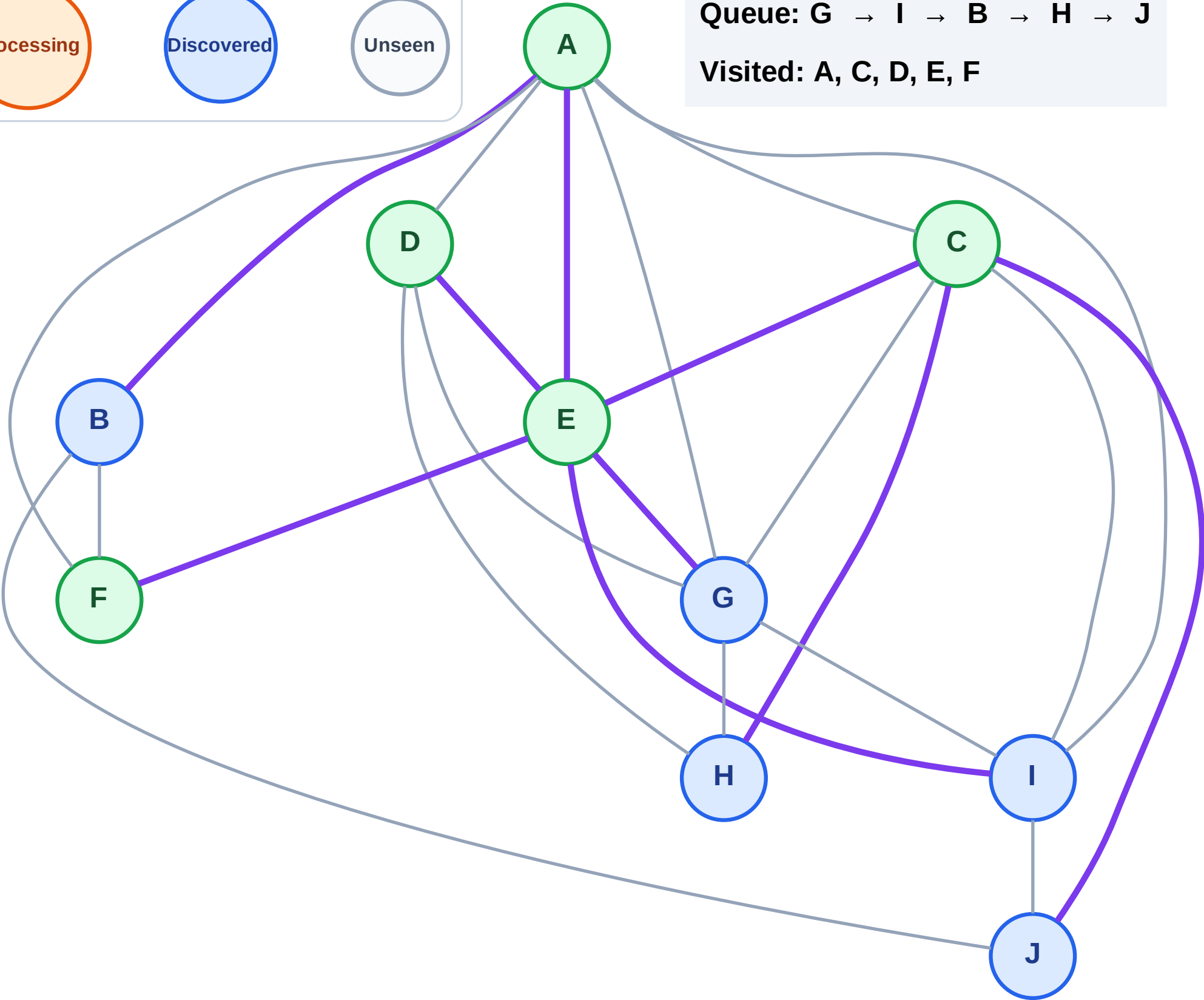
Processing

Discovered

Unseen

Queue: G → I → B → H → J

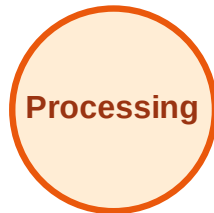
Visited: A, C, D, E, F



BFS Traversal

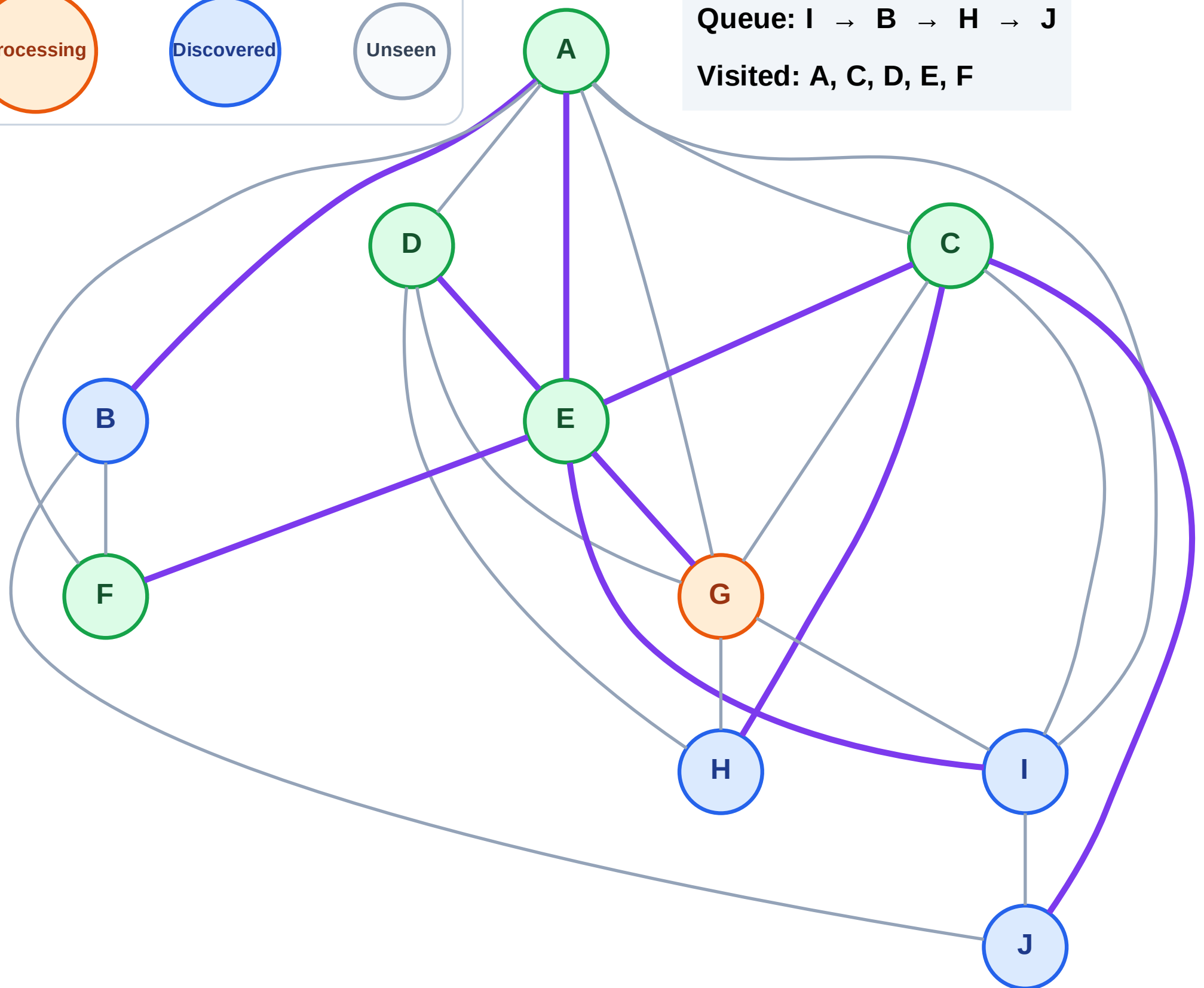
Step 12: Process node G

Legend



Queue: I → B → H → J

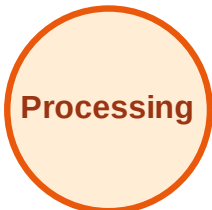
Visited: A, C, D, E, F



BFS Traversal

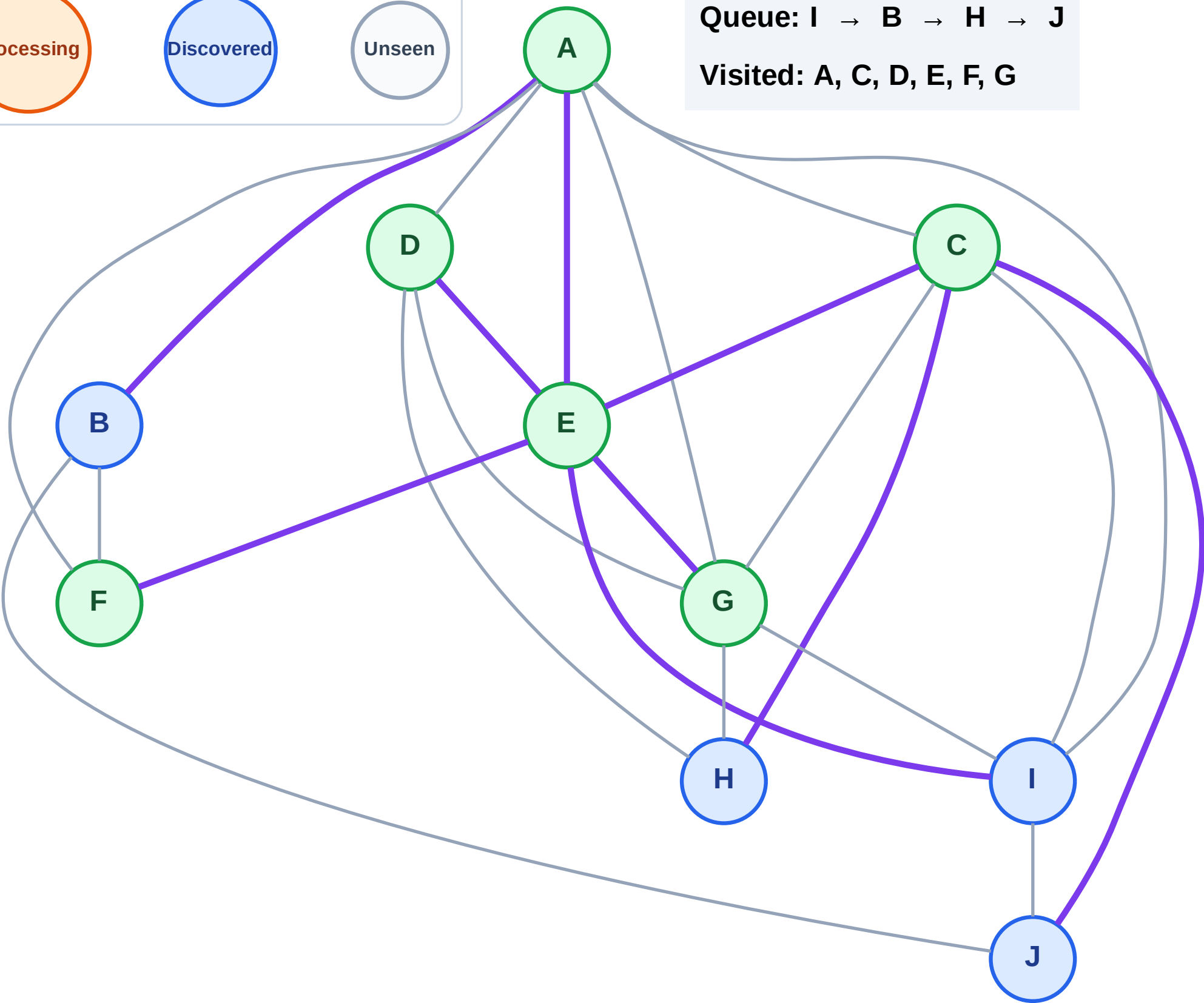
Step 13: No new neighbors from G

Legend



Queue: I → B → H → J

Visited: A, C, D, E, F, G



BFS Traversal

Step 14: Process node I

Legend

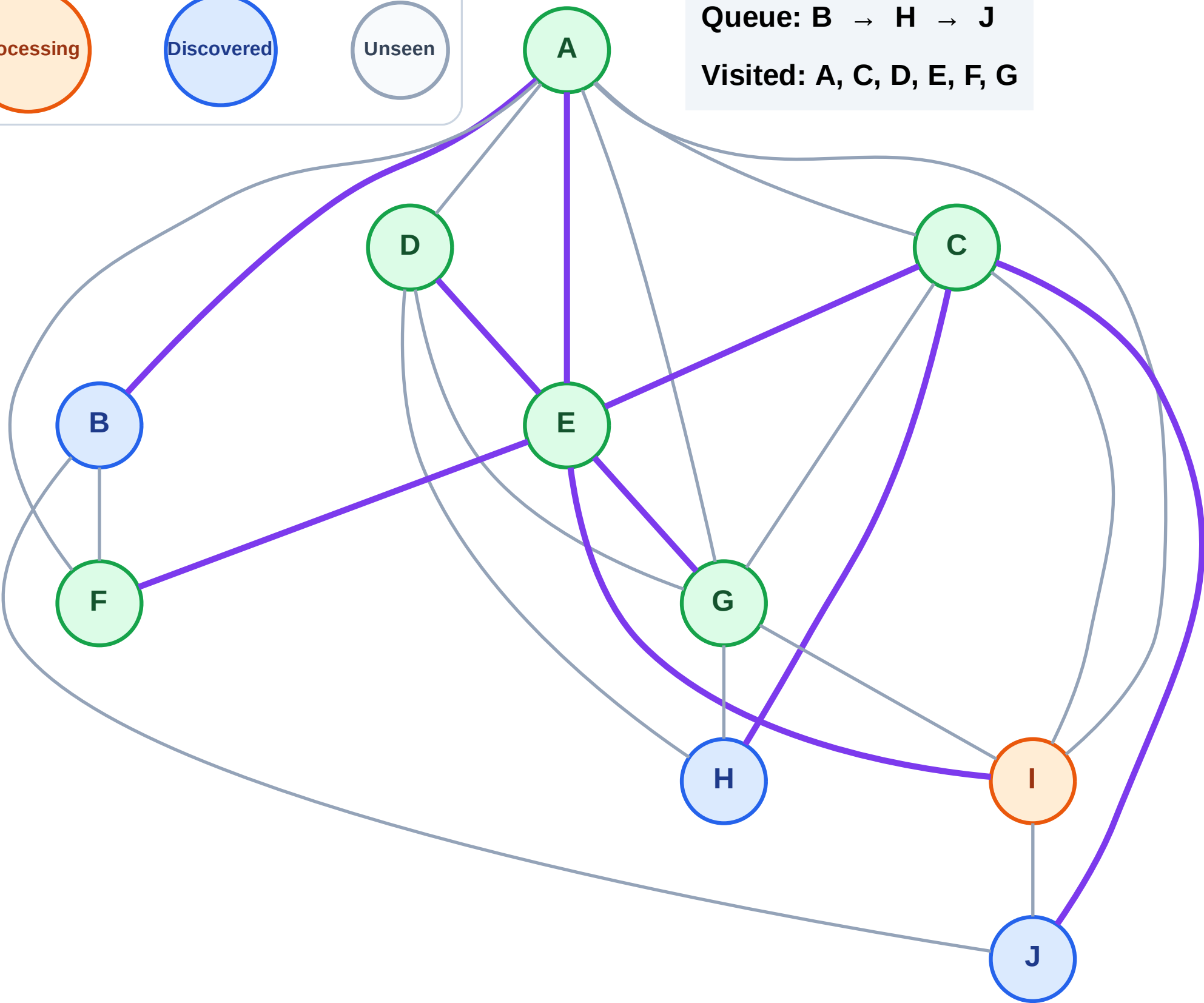
Complete

Processing

Discovered

Unseen

Queue: B → H → J
Visited: A, C, D, E, F, G



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

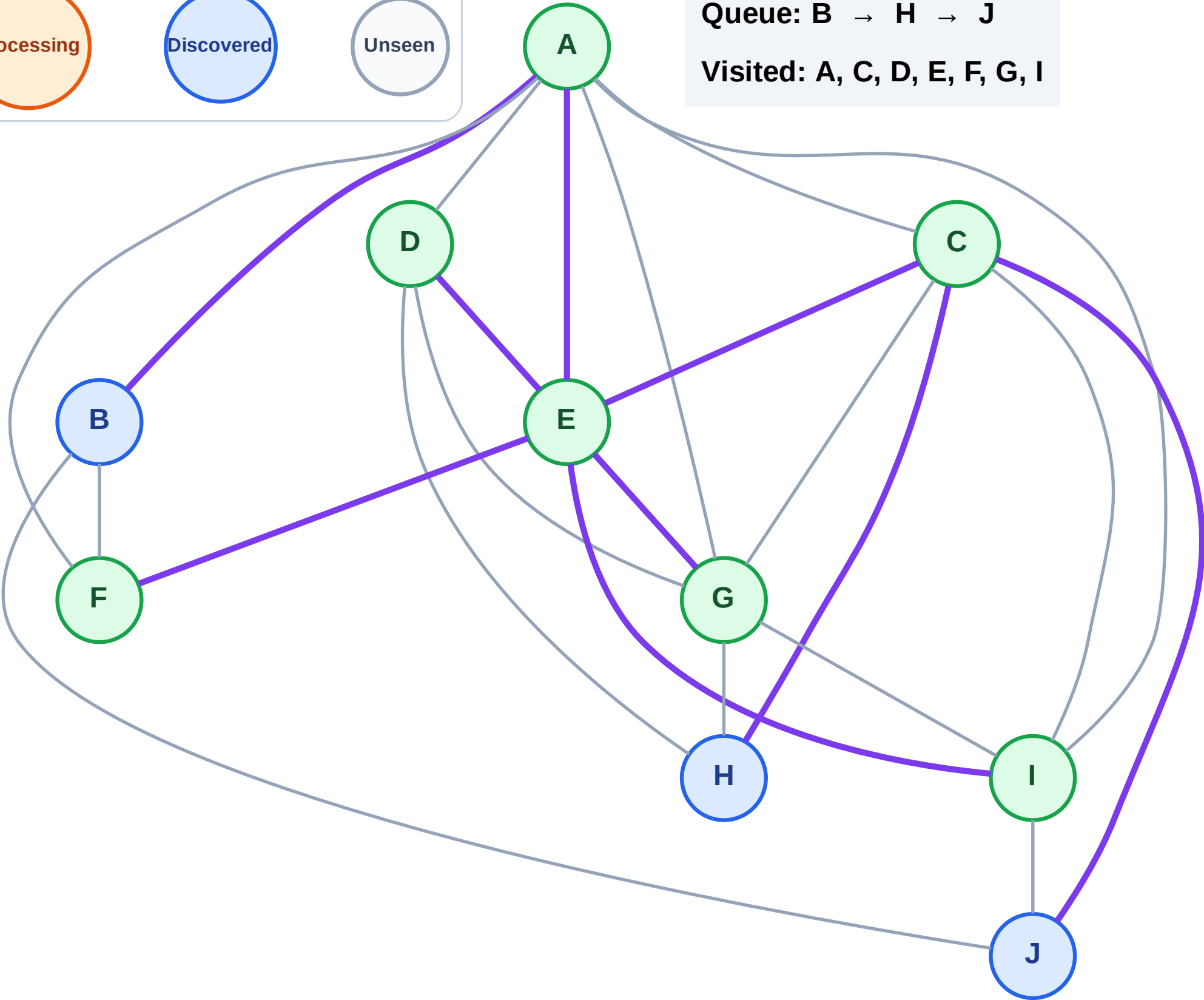
Processing

Discovered

Unseen

Queue: B → H → J

Visited: A, C, D, E, F, G, I



BFS Traversal

Step 16: Process node B

Legend

Complete

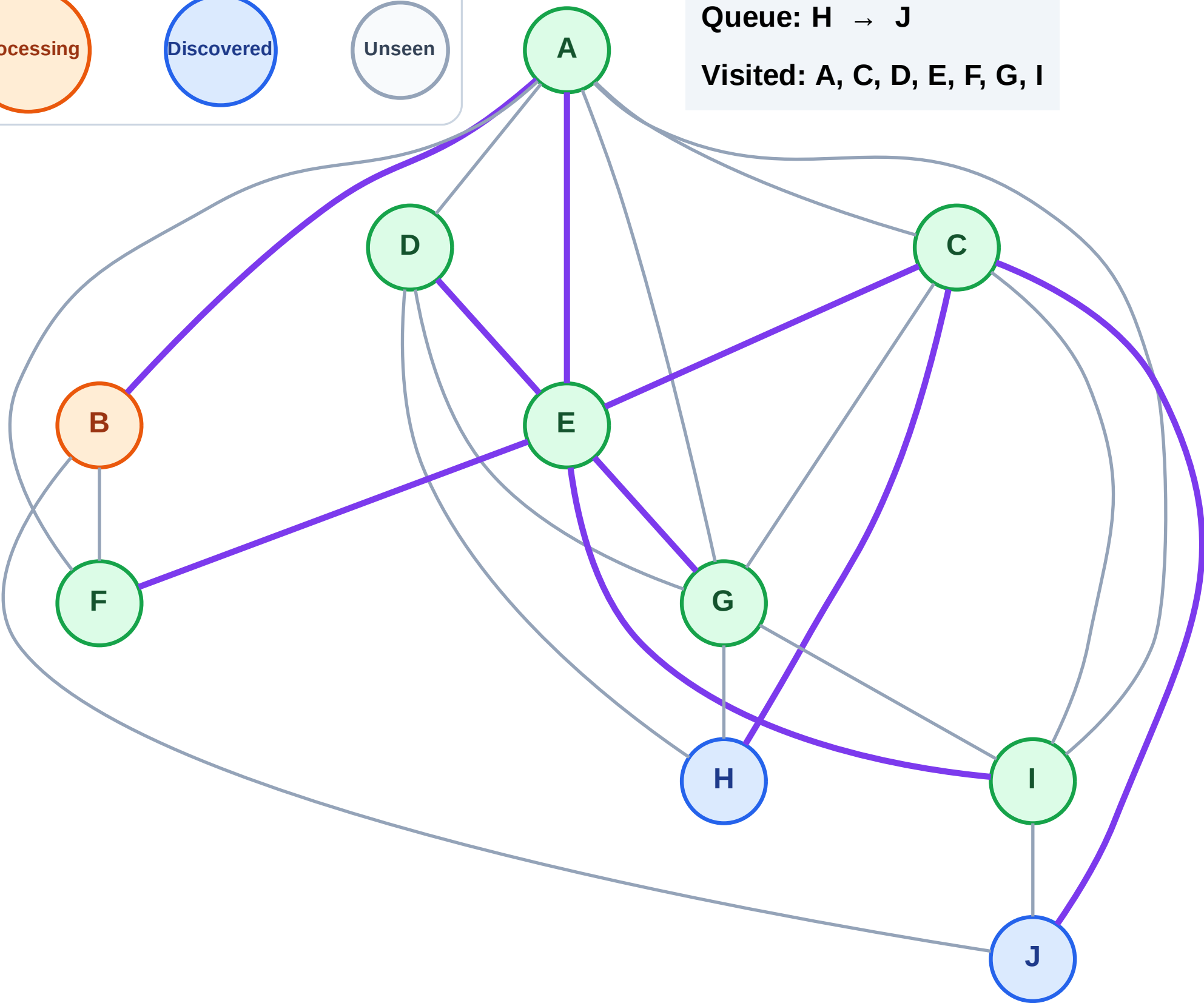
Processing

Discovered

Unseen

Queue: H → J

Visited: A, C, D, E, F, G, I



BFS Traversal

Step 17: No new neighbors from B

Legend

Complete

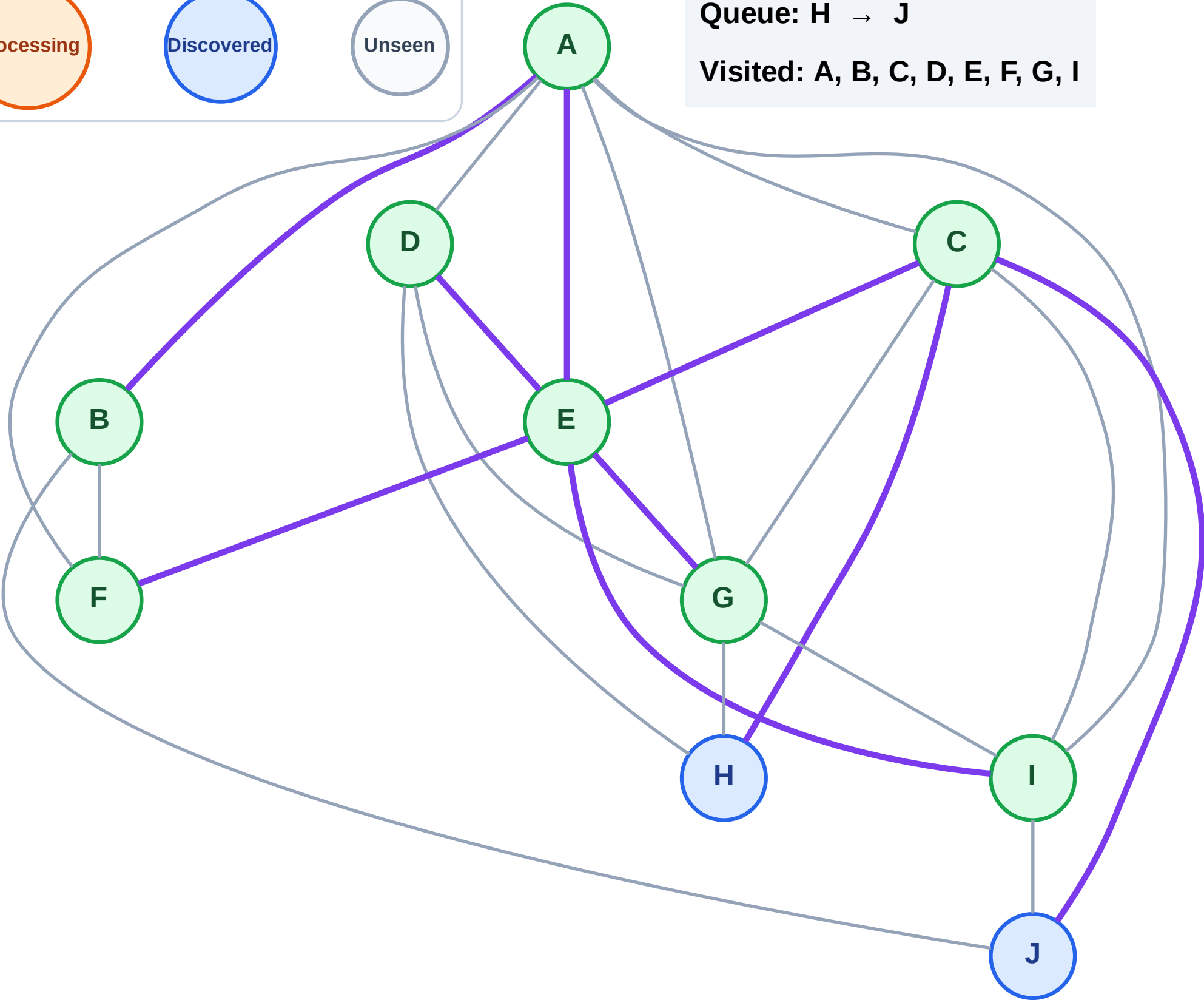
Processing

Discovered

Unseen

Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

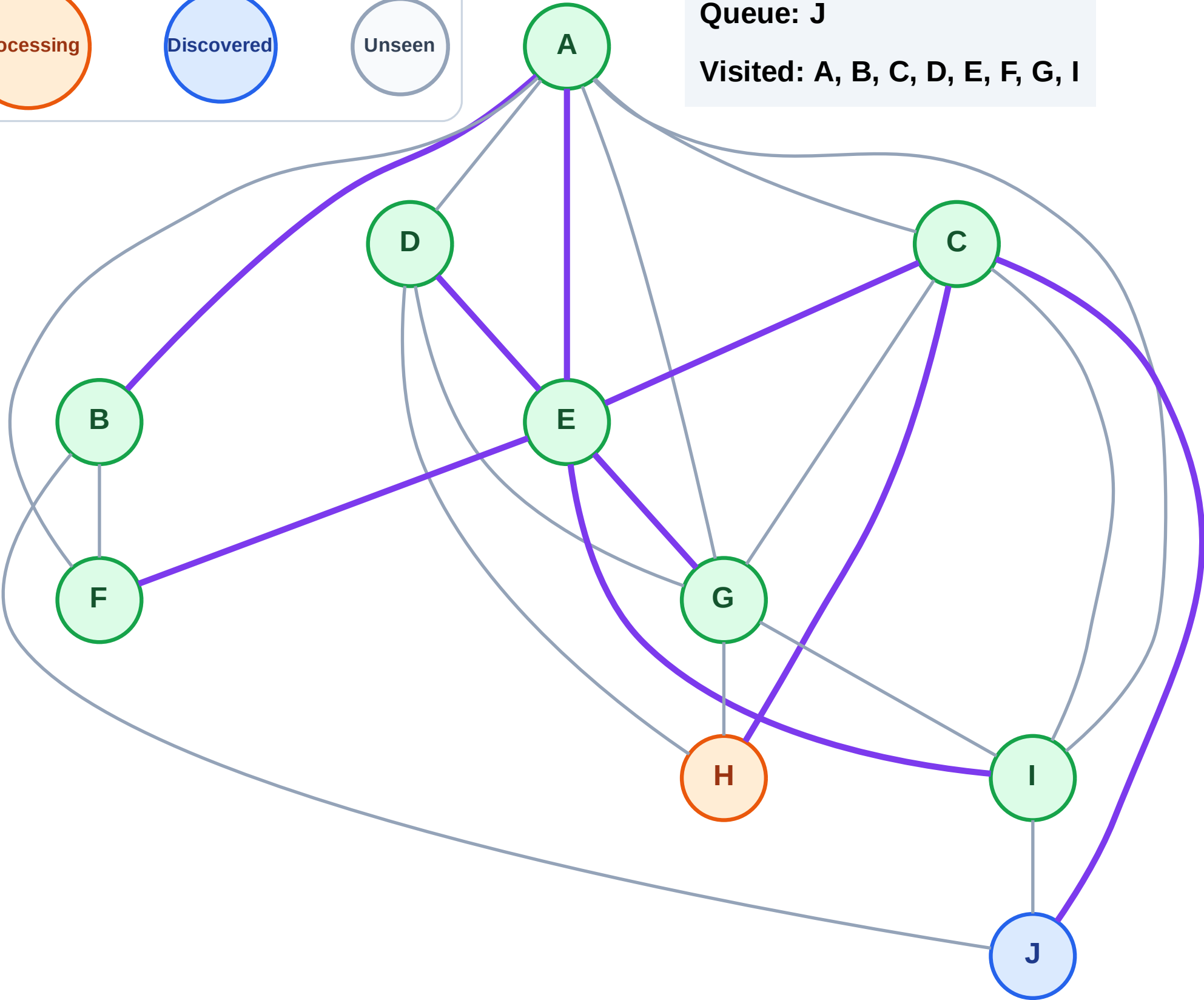
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 19: No new neighbors from H

Legend

Complete

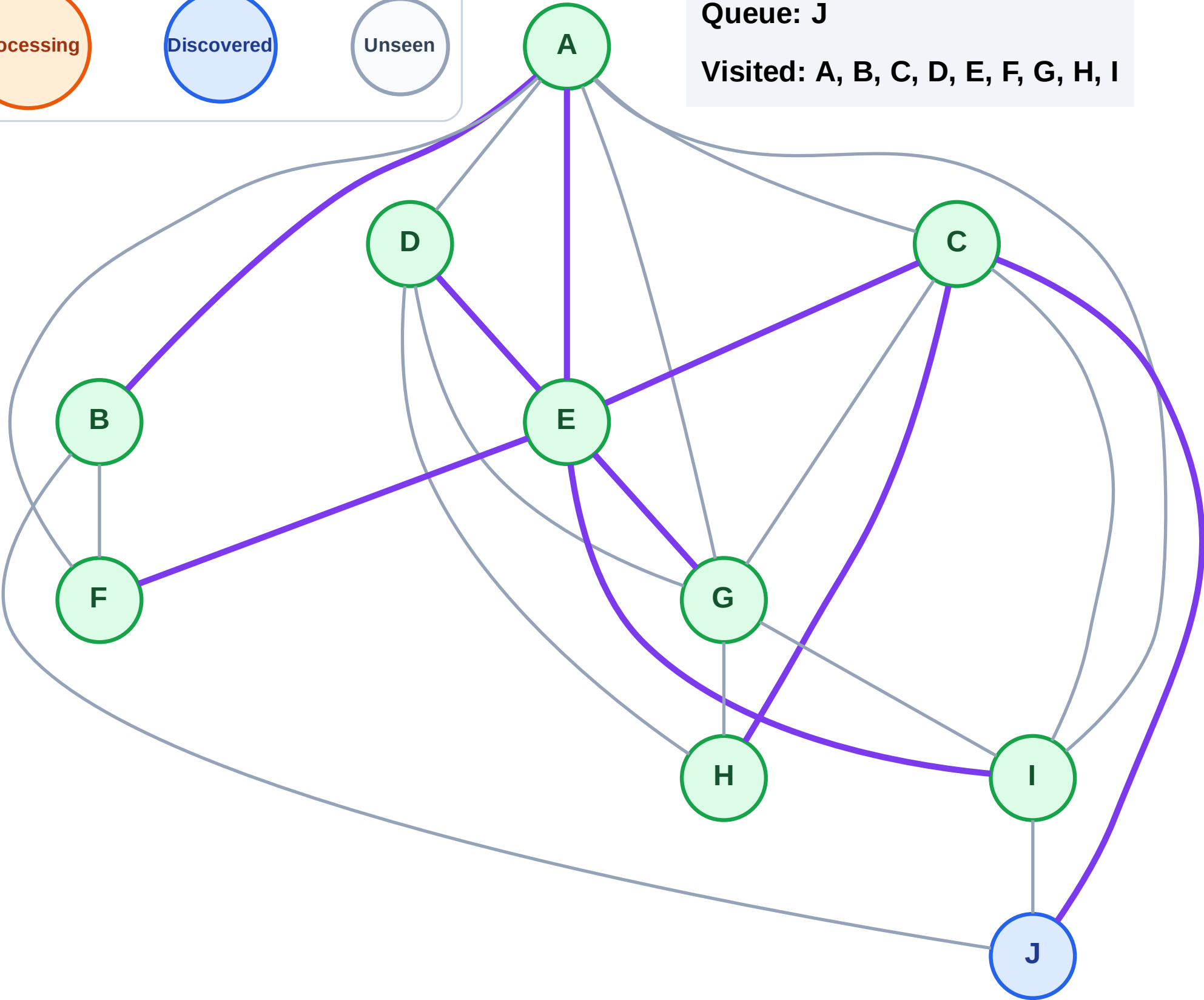
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

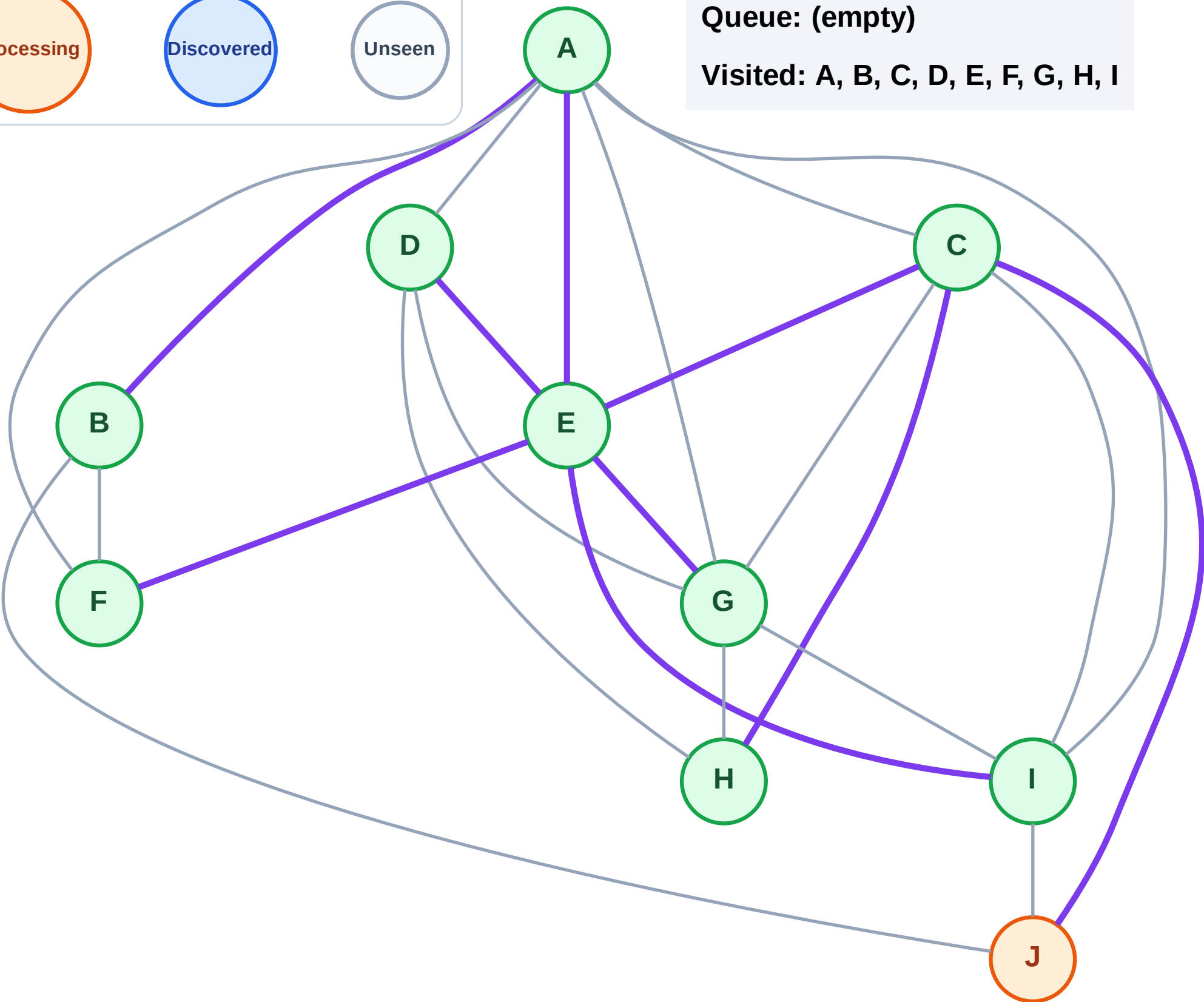
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 21: No new neighbors from J

Legend

Complete

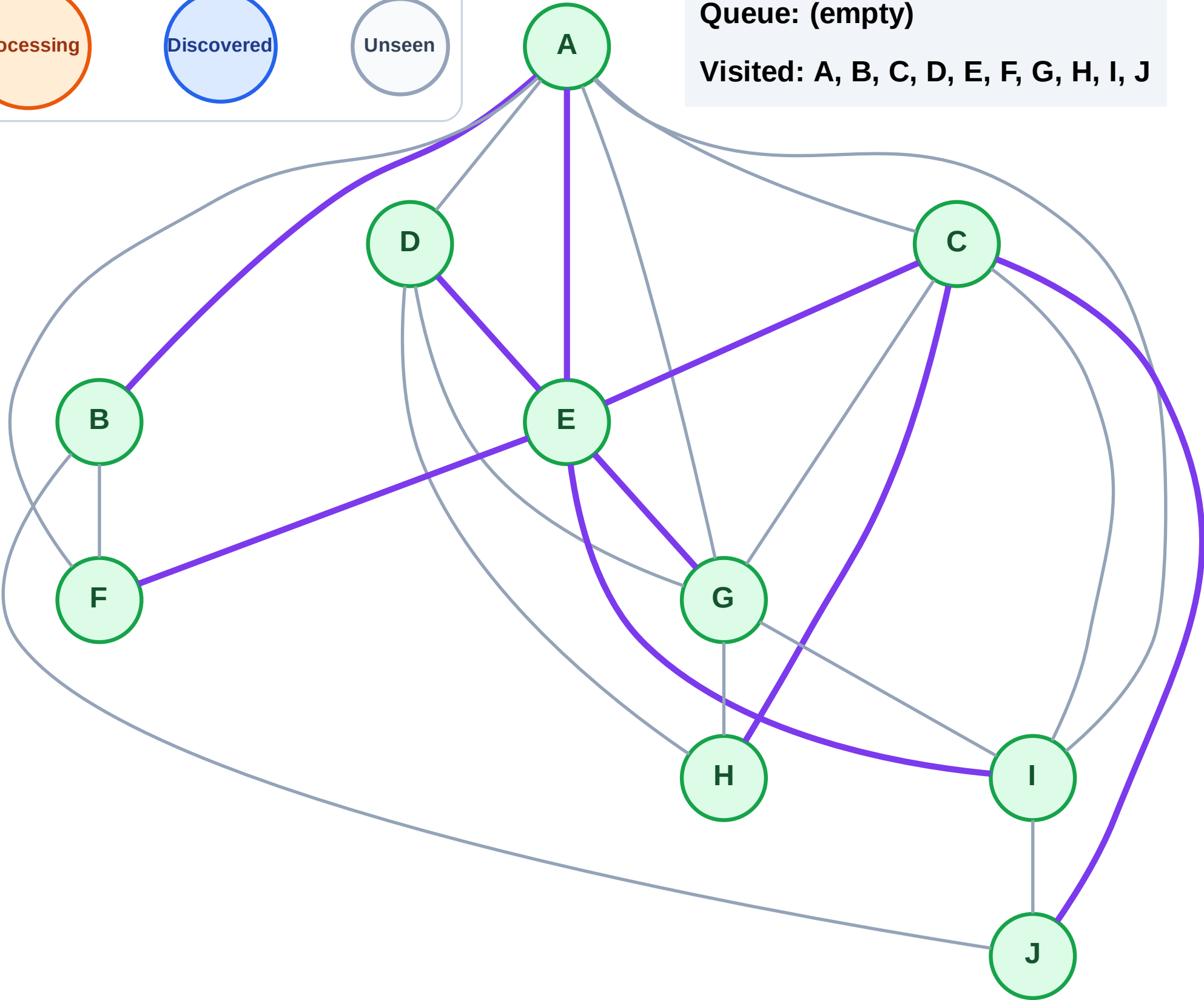
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

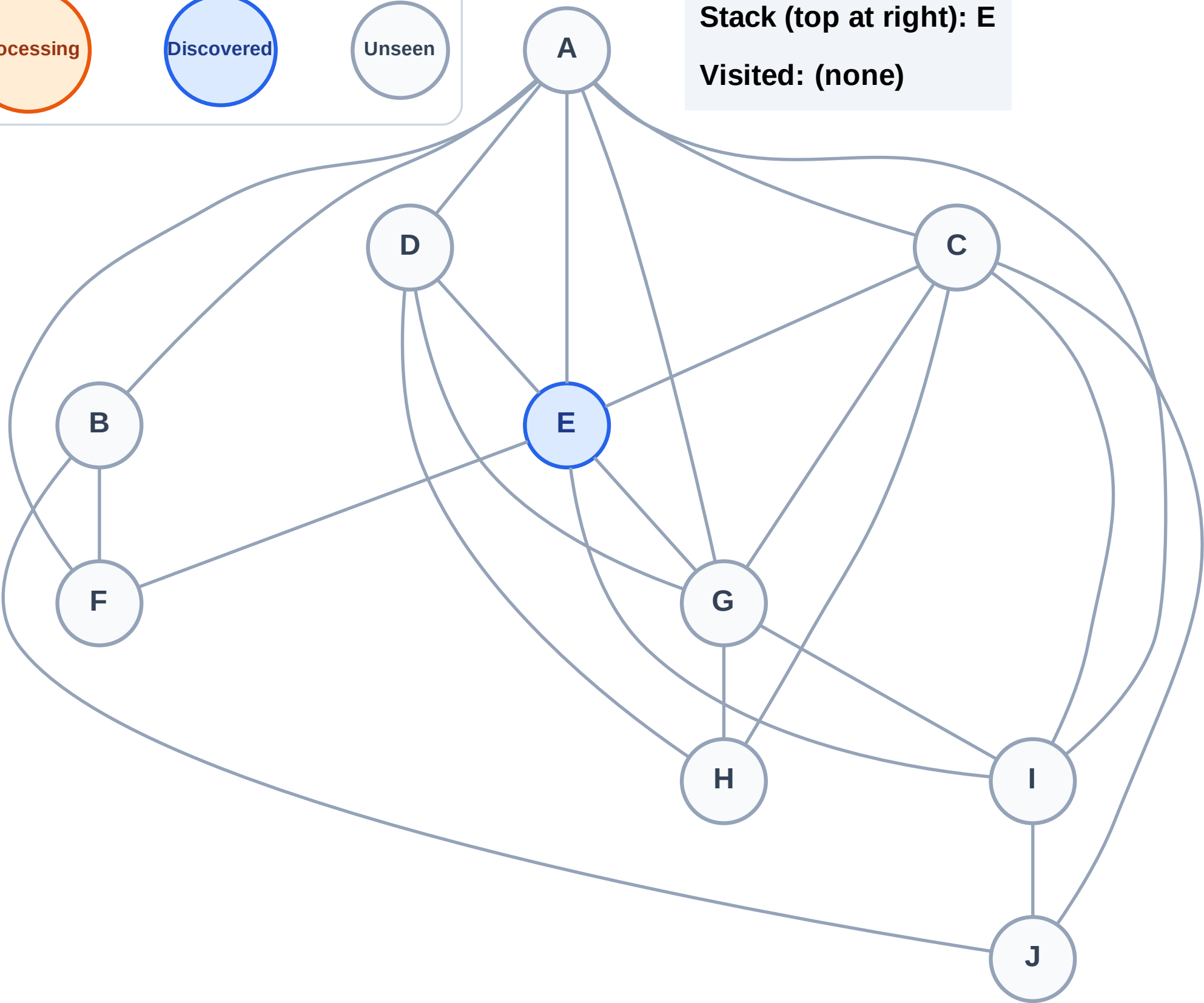
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

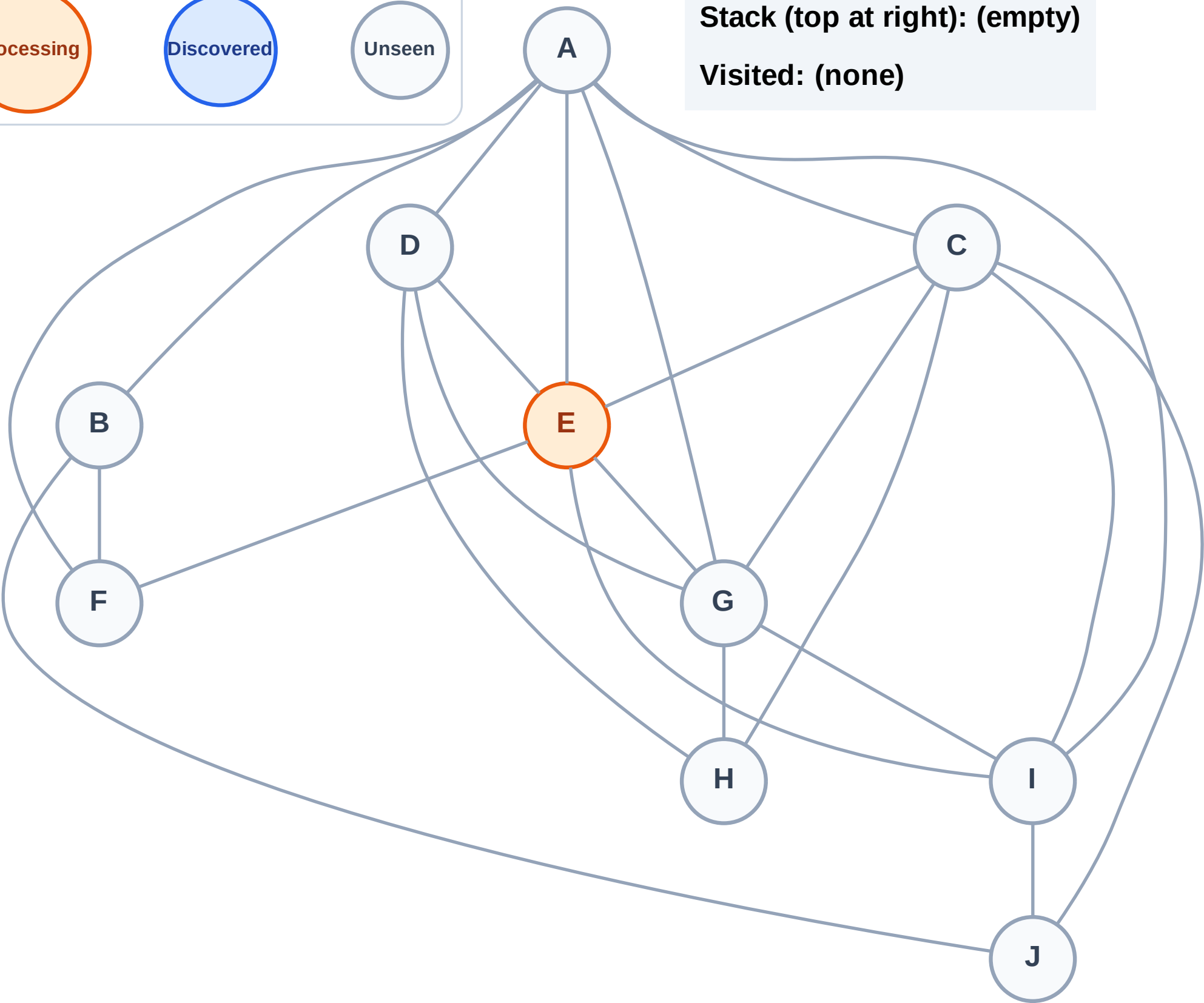
Step 2: Process node E

Legend



Stack (top at right): (empty)

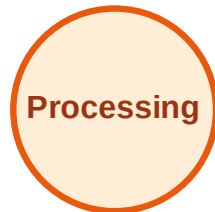
Visited: (none)



DFS Traversal

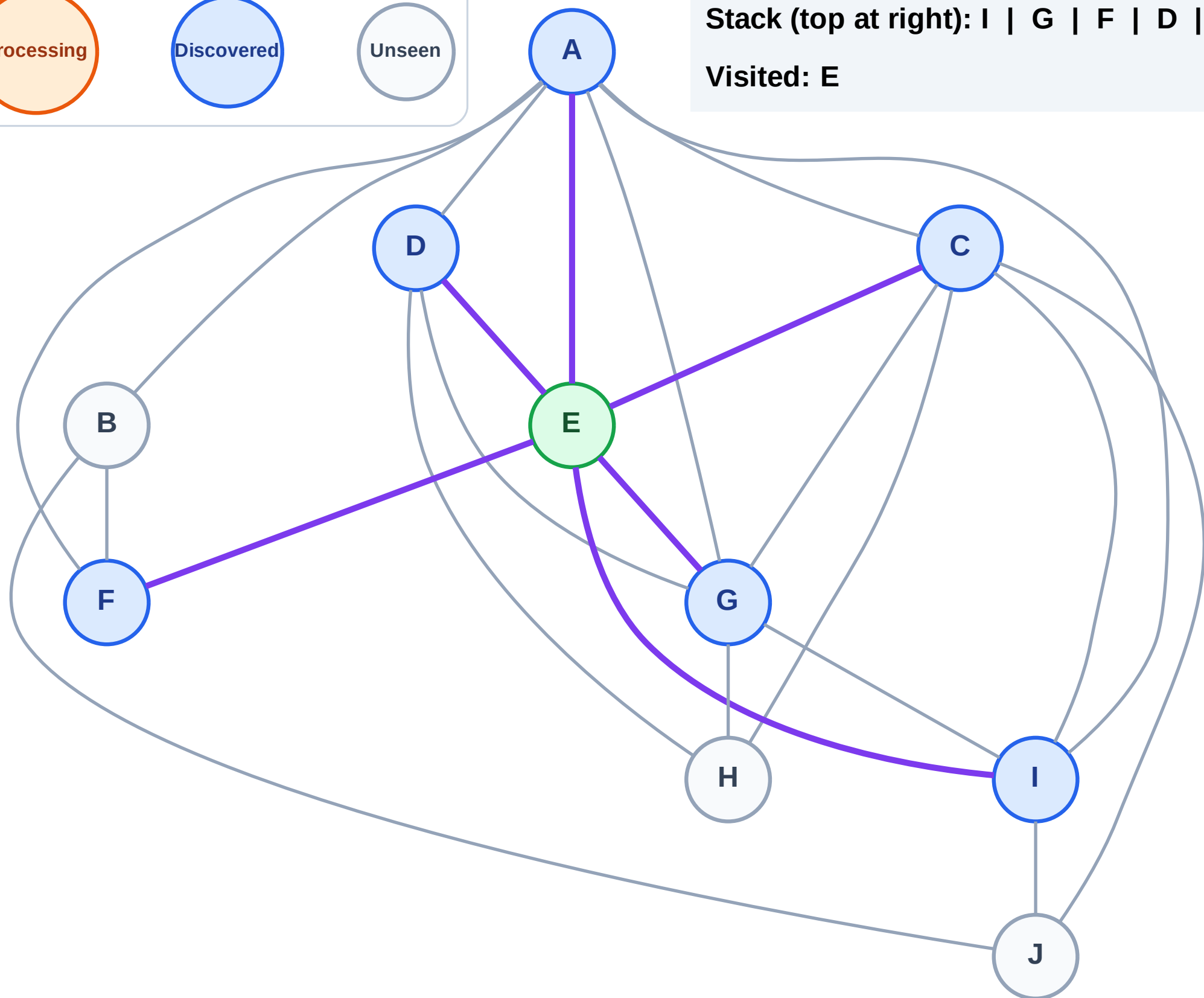
Step 3: Discover I, G, F, D, C, A from E

Legend



Stack (top at right): I | G | F | D | C | A

Visited: E



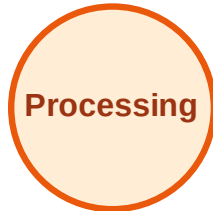
DFS Traversal

Step 4: Process node A

Legend



Complete



Processing



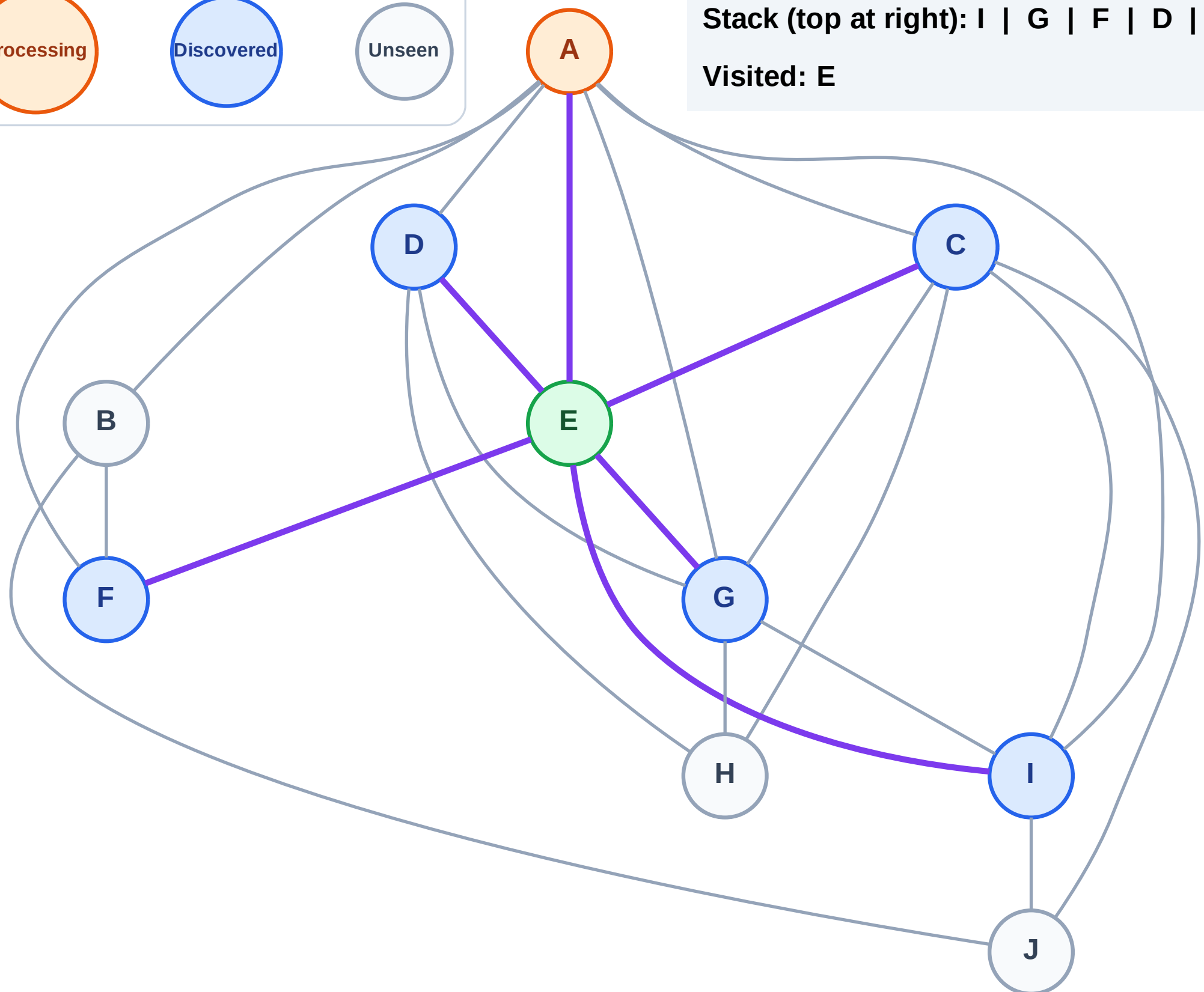
Discovered



Unseen

Stack (top at right): I | G | F | D | C

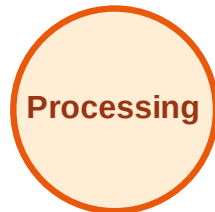
Visited: E



DFS Traversal

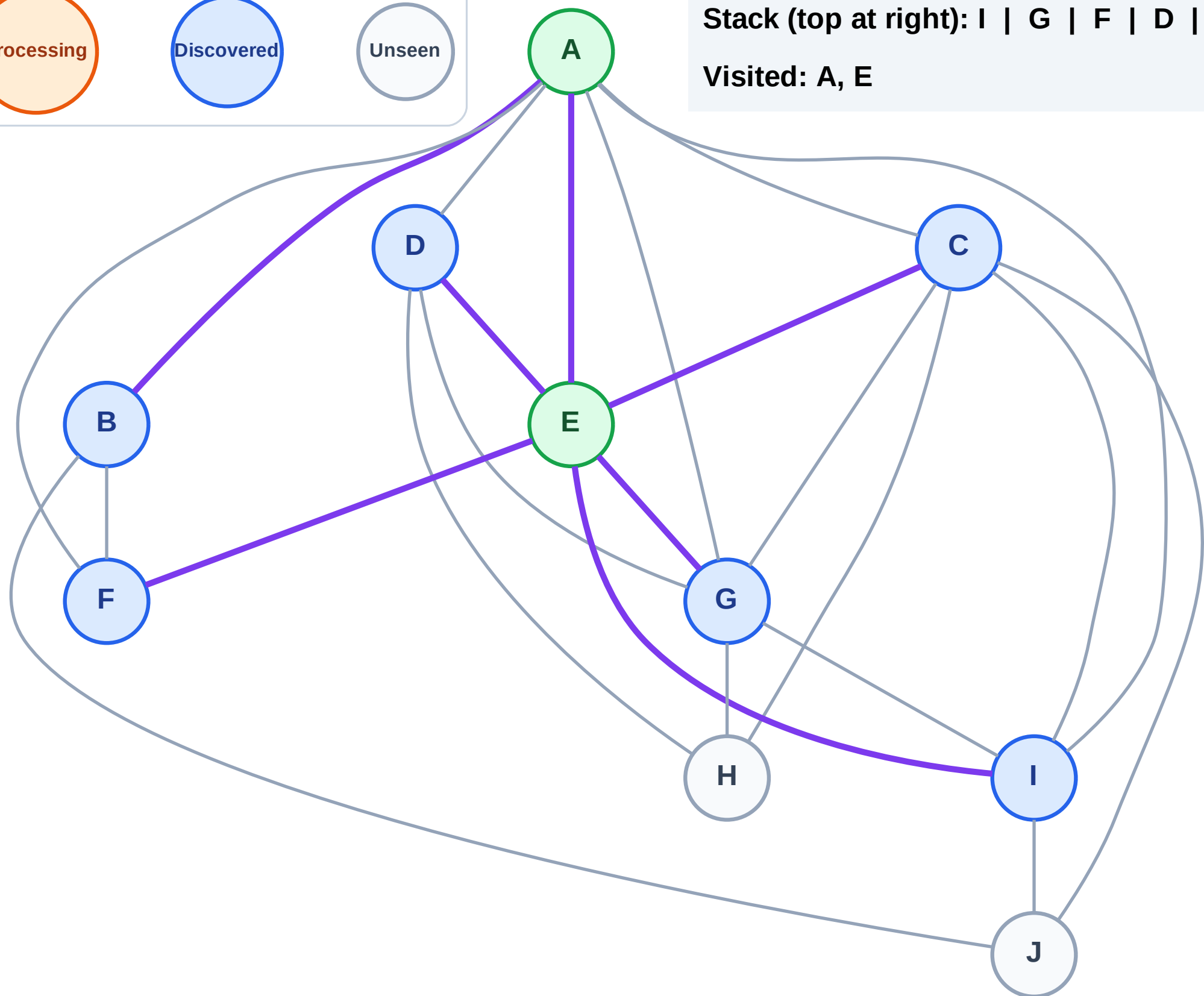
Step 5: Discover B from A

Legend



Stack (top at right): I | G | F | D | C | B

Visited: A, E



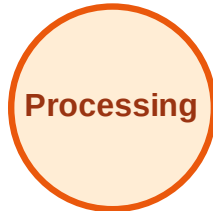
DFS Traversal

Step 6: Process node B

Legend



Complete



Processing



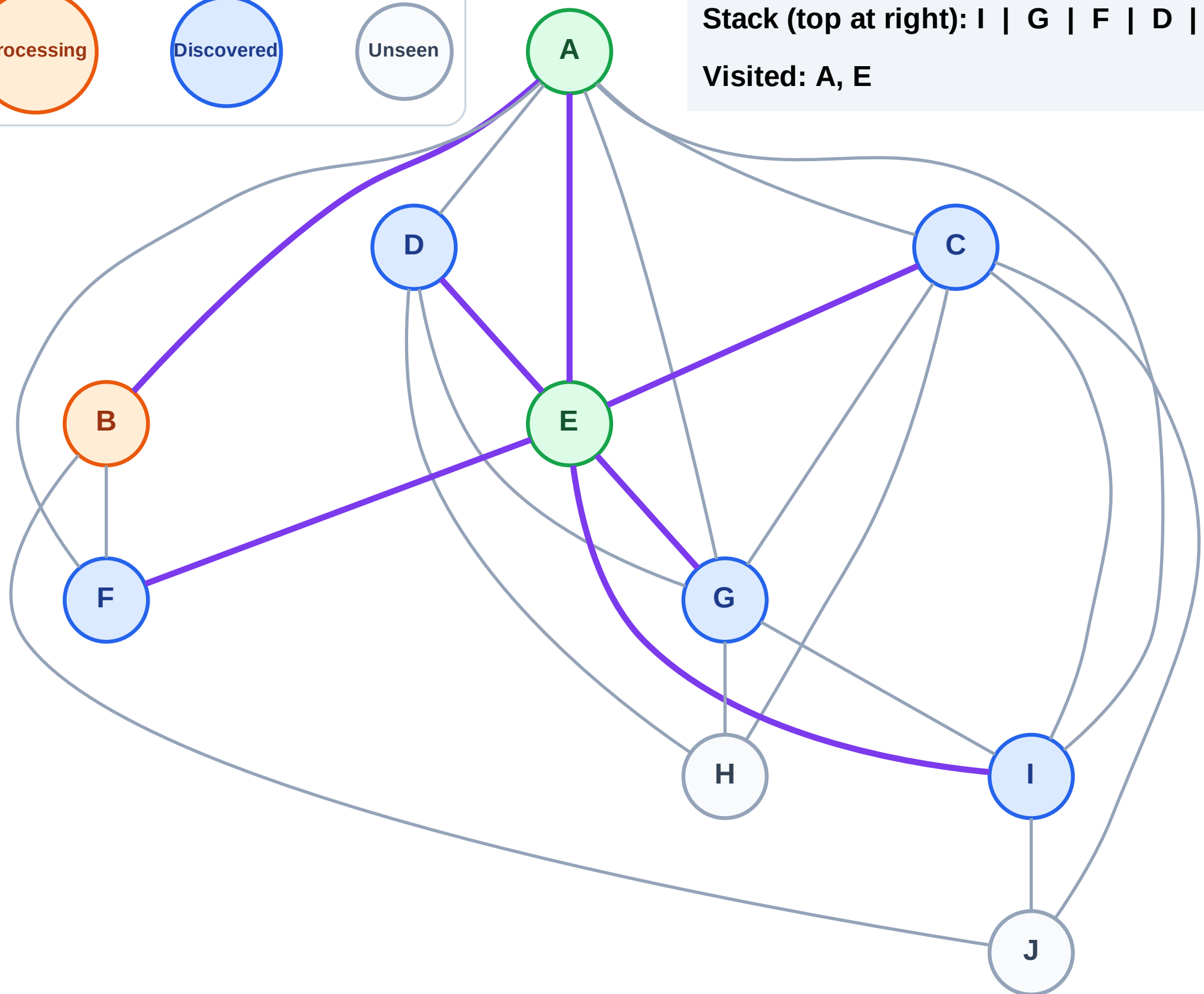
Discovered



Unseen

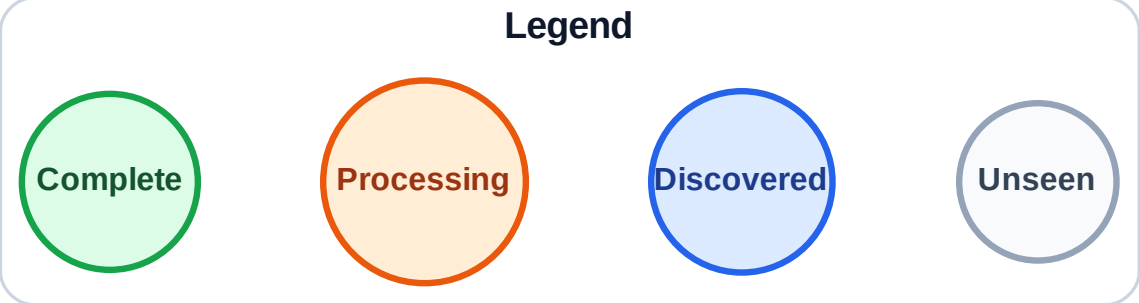
Stack (top at right): I | G | F | D | C

Visited: A, E



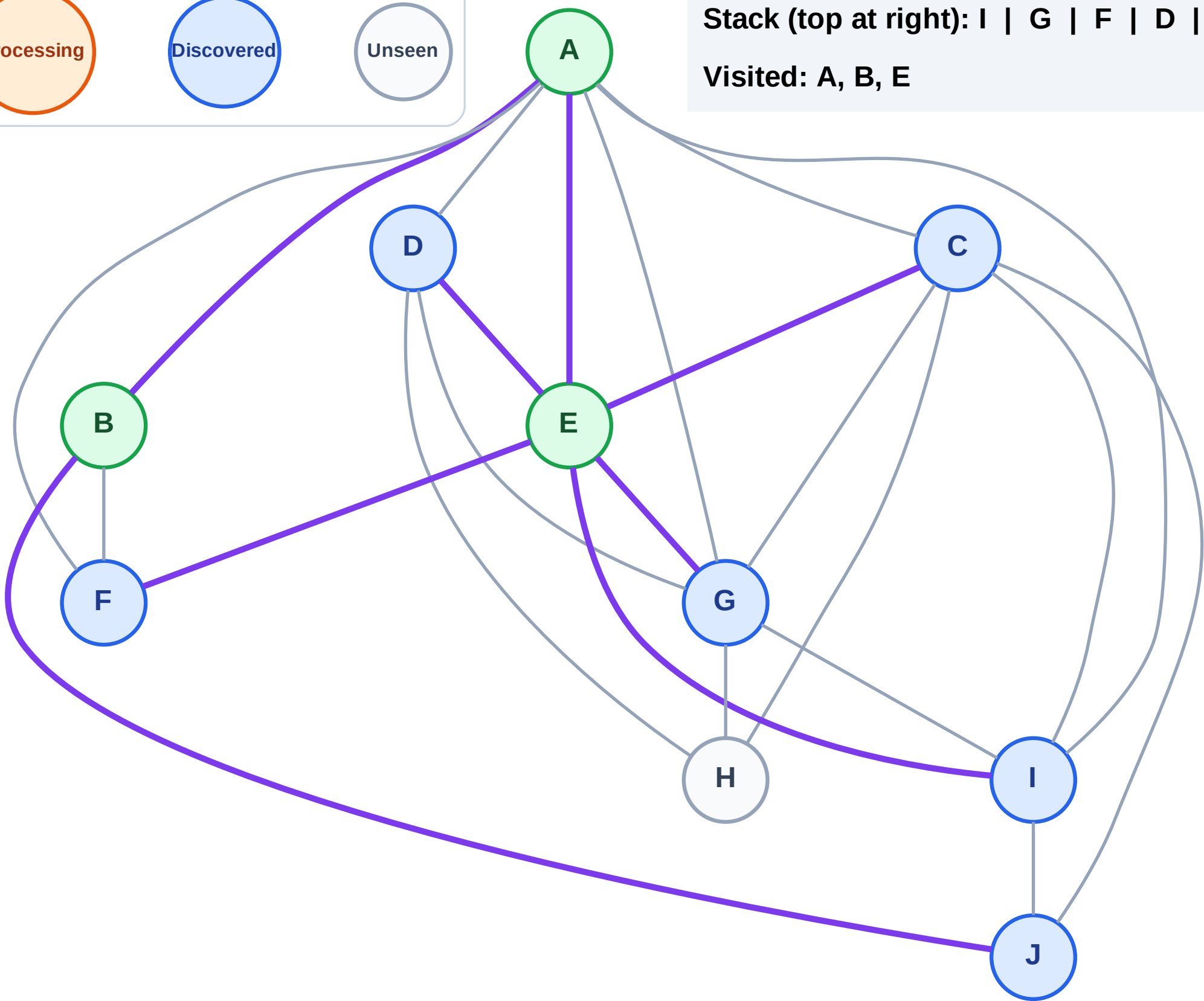
Step 7: Discover J from B

Figure 1. The effect of the number of nodes on the number of nodes in the network.



Stack (top at right): I | G | F | D | C | J

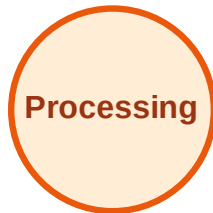
Visited: A, B, E



DFS Traversal

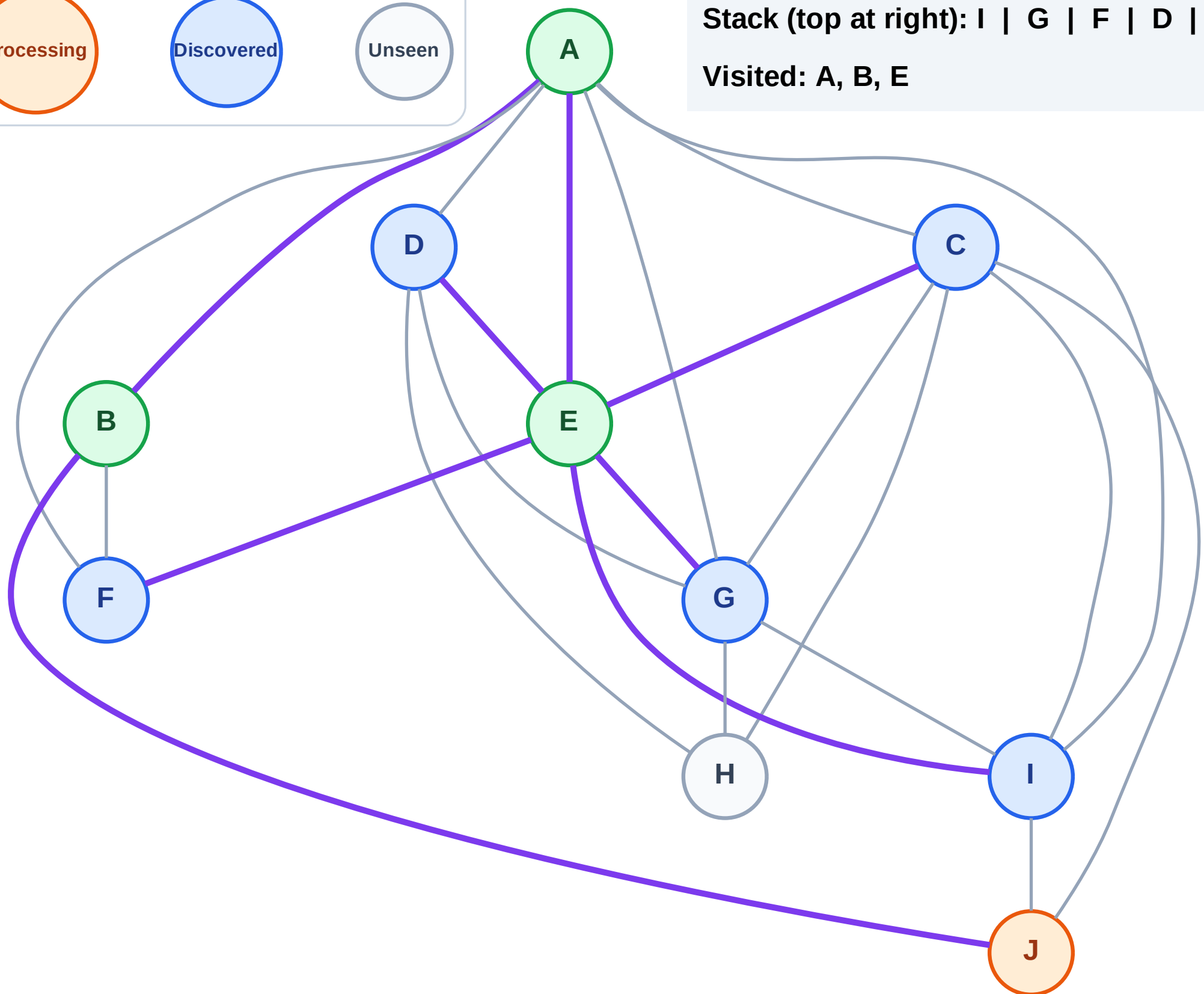
Step 8: Process node J

Legend



Stack (top at right): I | G | F | D | C

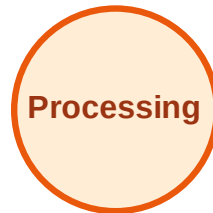
Visited: A, B, E



DFS Traversal

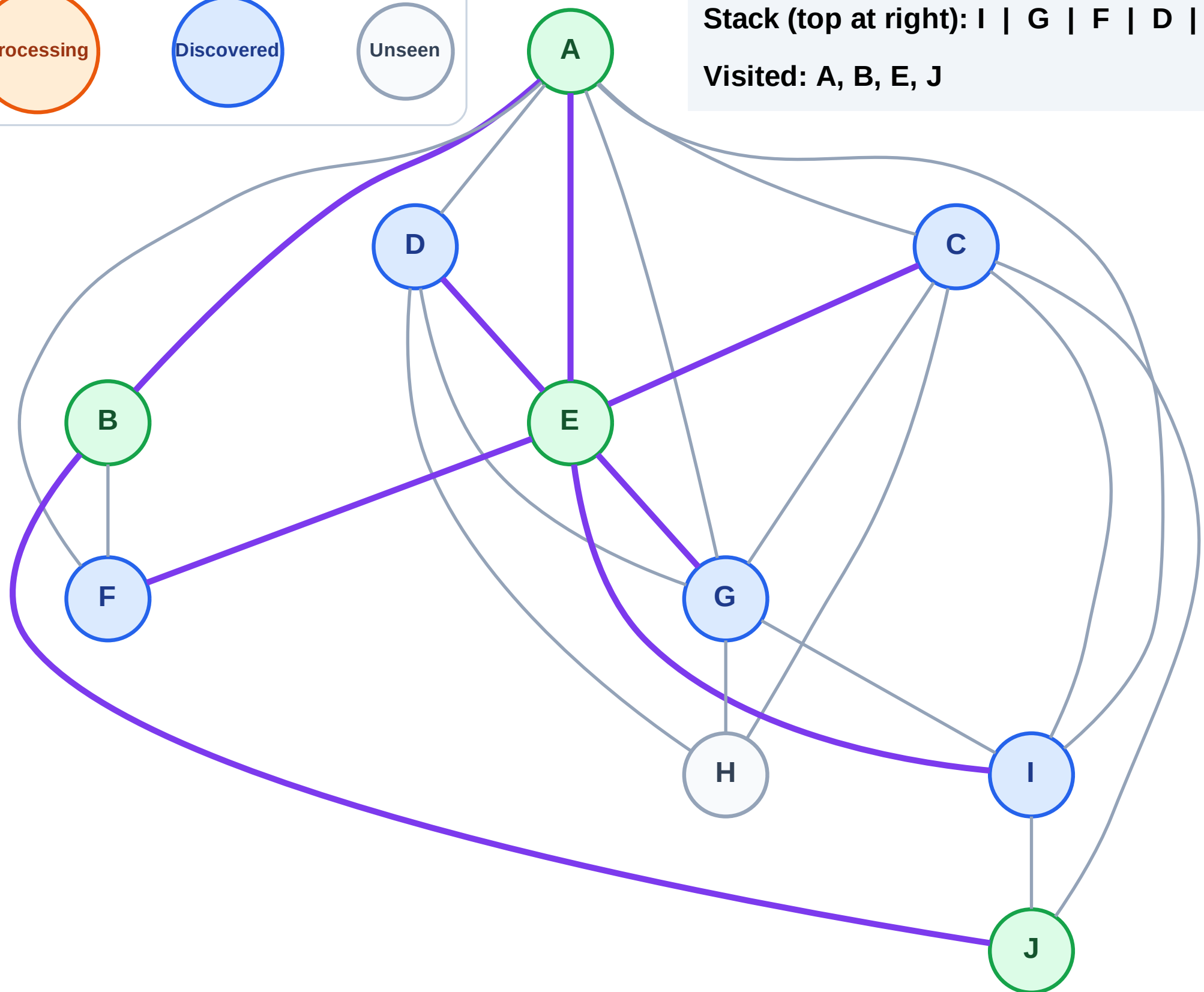
Step 9: No new neighbors from J

Legend



Stack (top at right): I | G | F | D | C

Visited: A, B, E, J



DFS Traversal

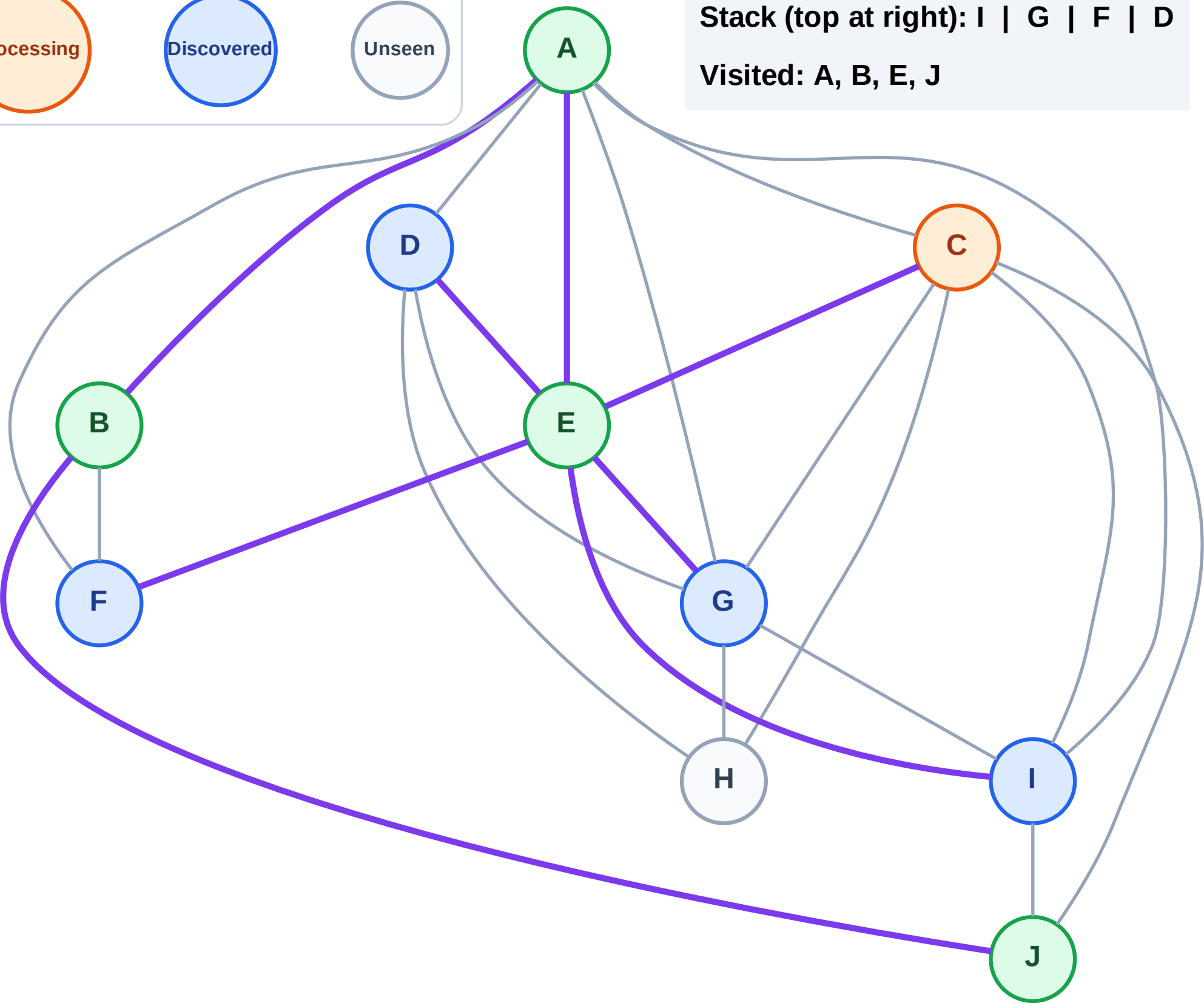
Step 10: Process node C

Legend



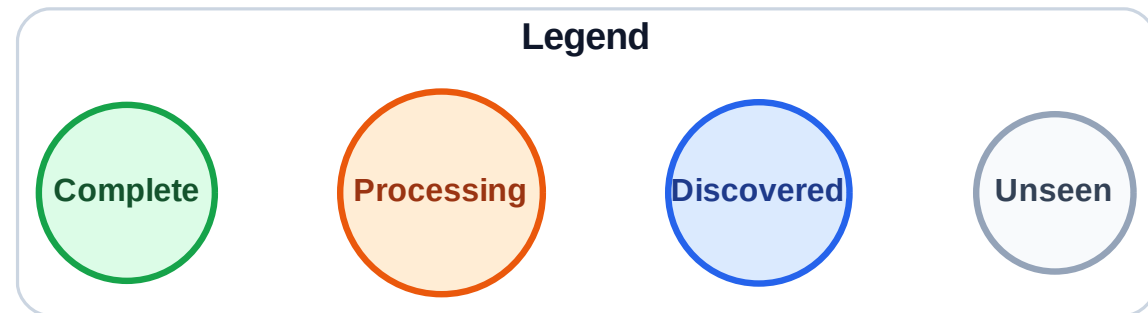
Stack (top at right): I | G | F | D

Visited: A, B, E, J



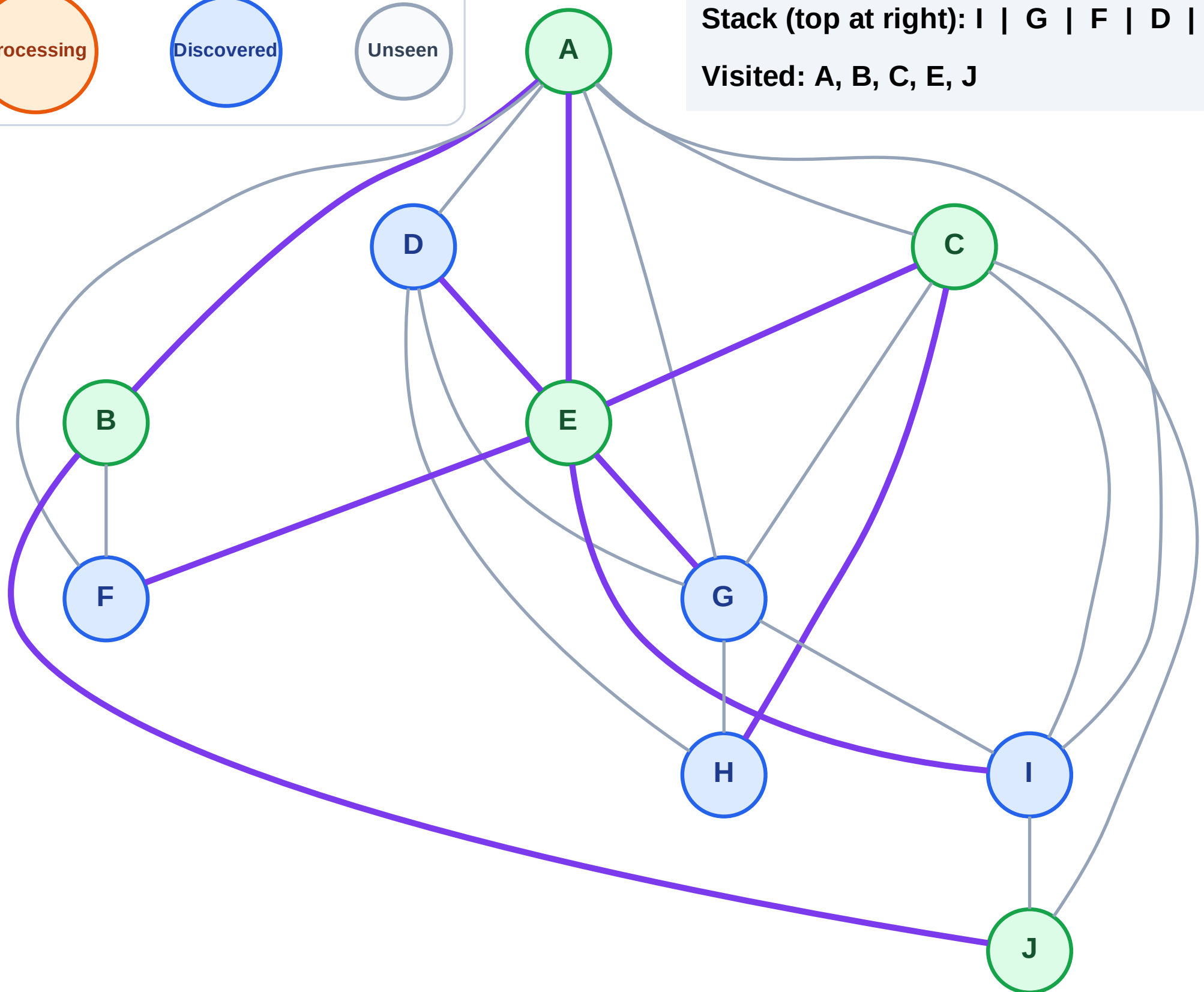
DFS Traversal

Step 11: Discover H from C



Stack (top at right): I | G | F | D | H

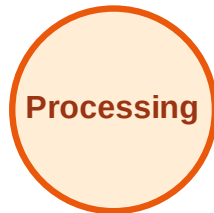
Visited: A, B, C, E, J



DFS Traversal

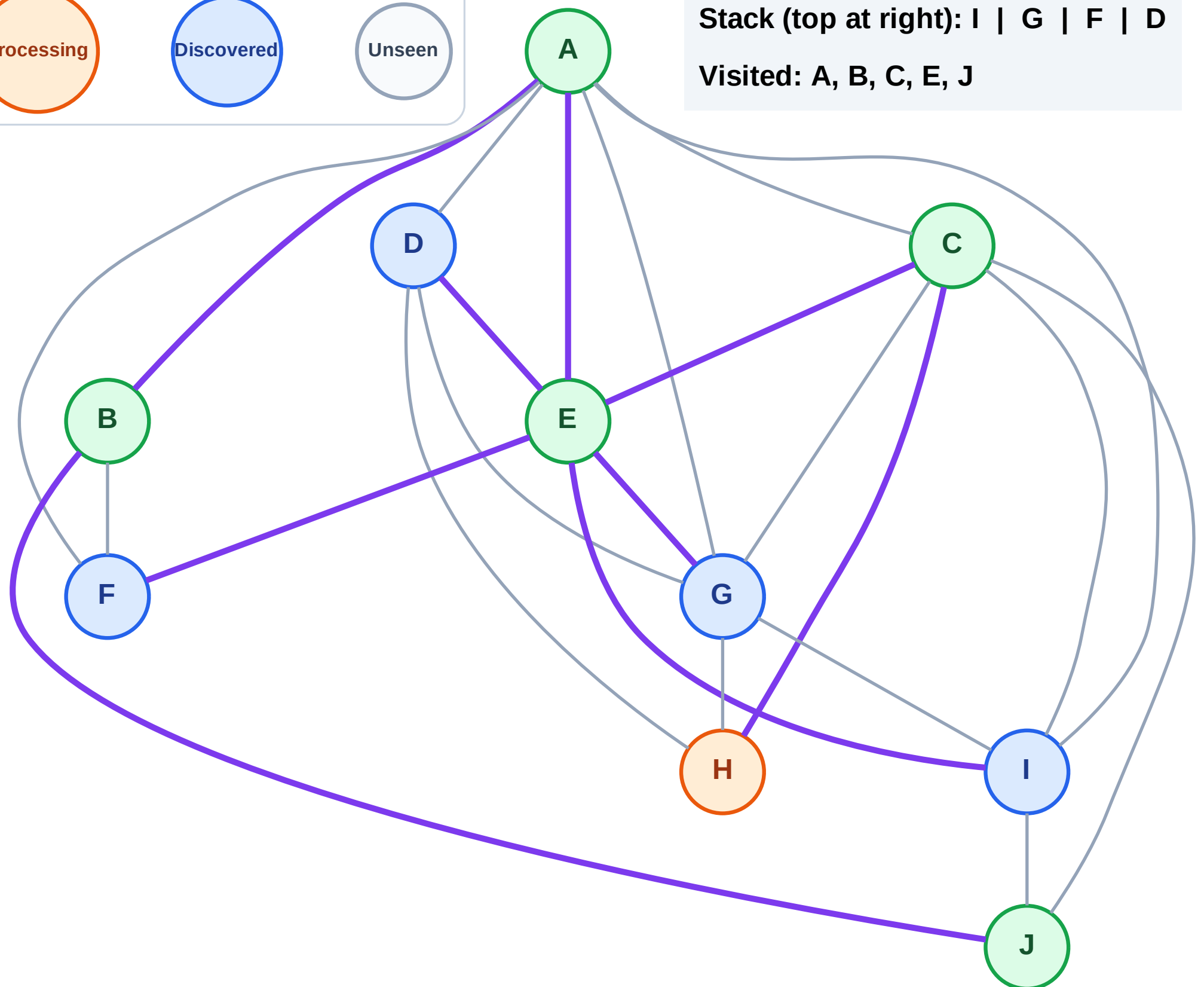
Step 12: Process node H

Legend



Stack (top at right): I | G | F | D

Visited: A, B, C, E, J



DFS Traversal

Step 13: No new neighbors from H

Legend

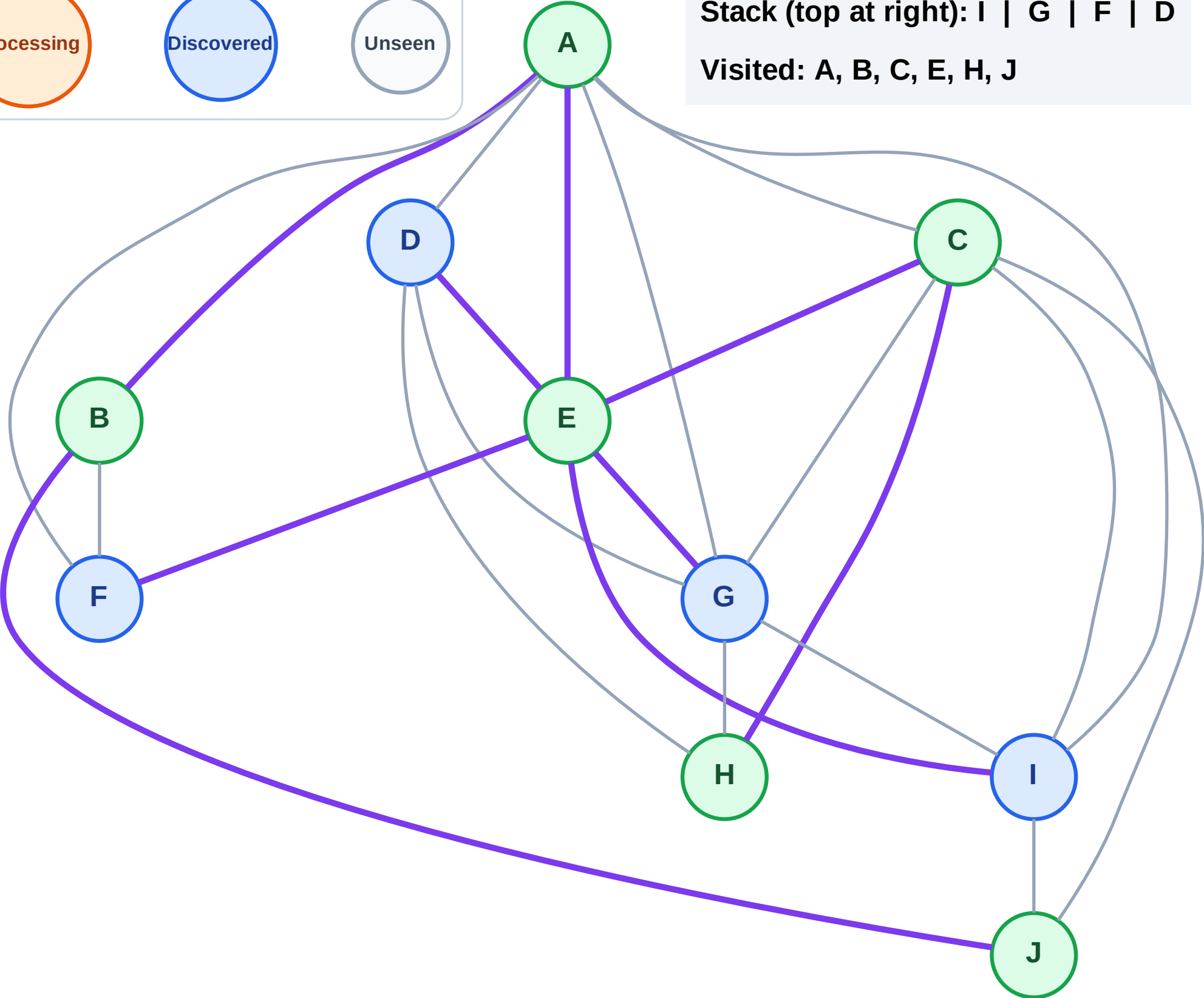
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | F | D
Visited: A, B, C, E, H, J



DFS Traversal

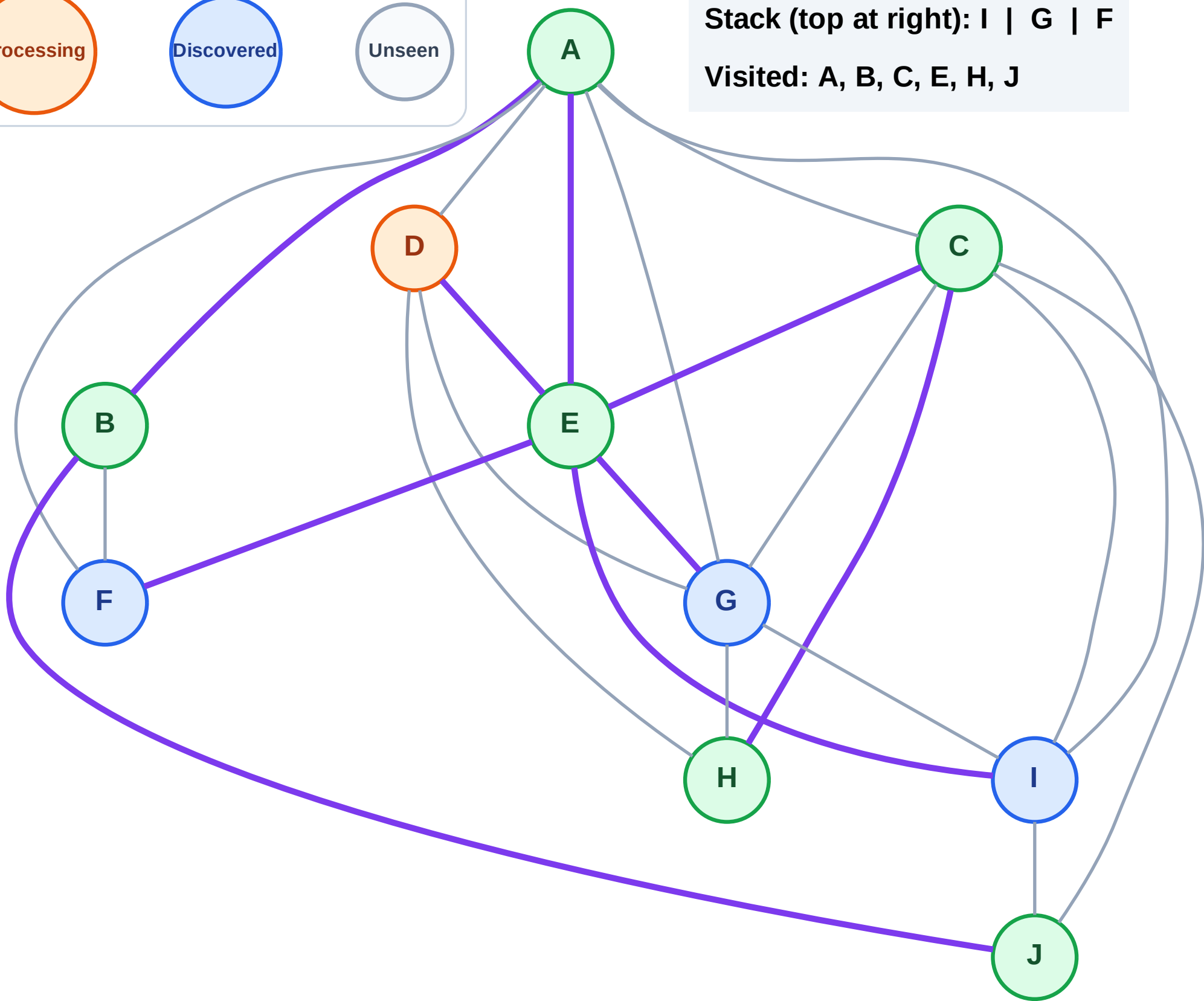
Step 14: Process node D

Legend



Stack (top at right): I | G | F

Visited: A, B, C, E, H, J



DFS Traversal

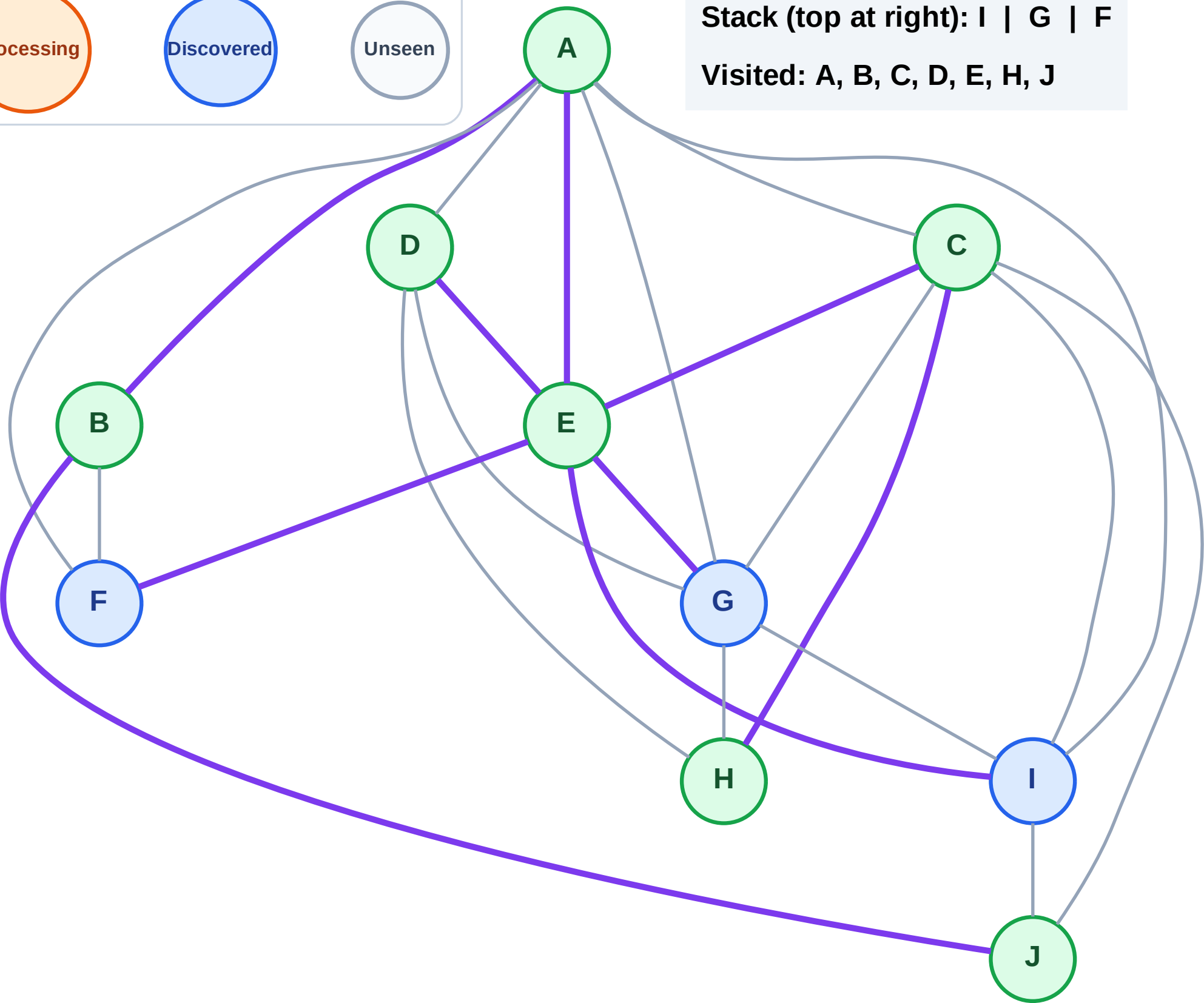
Step 15: No new neighbors from D

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

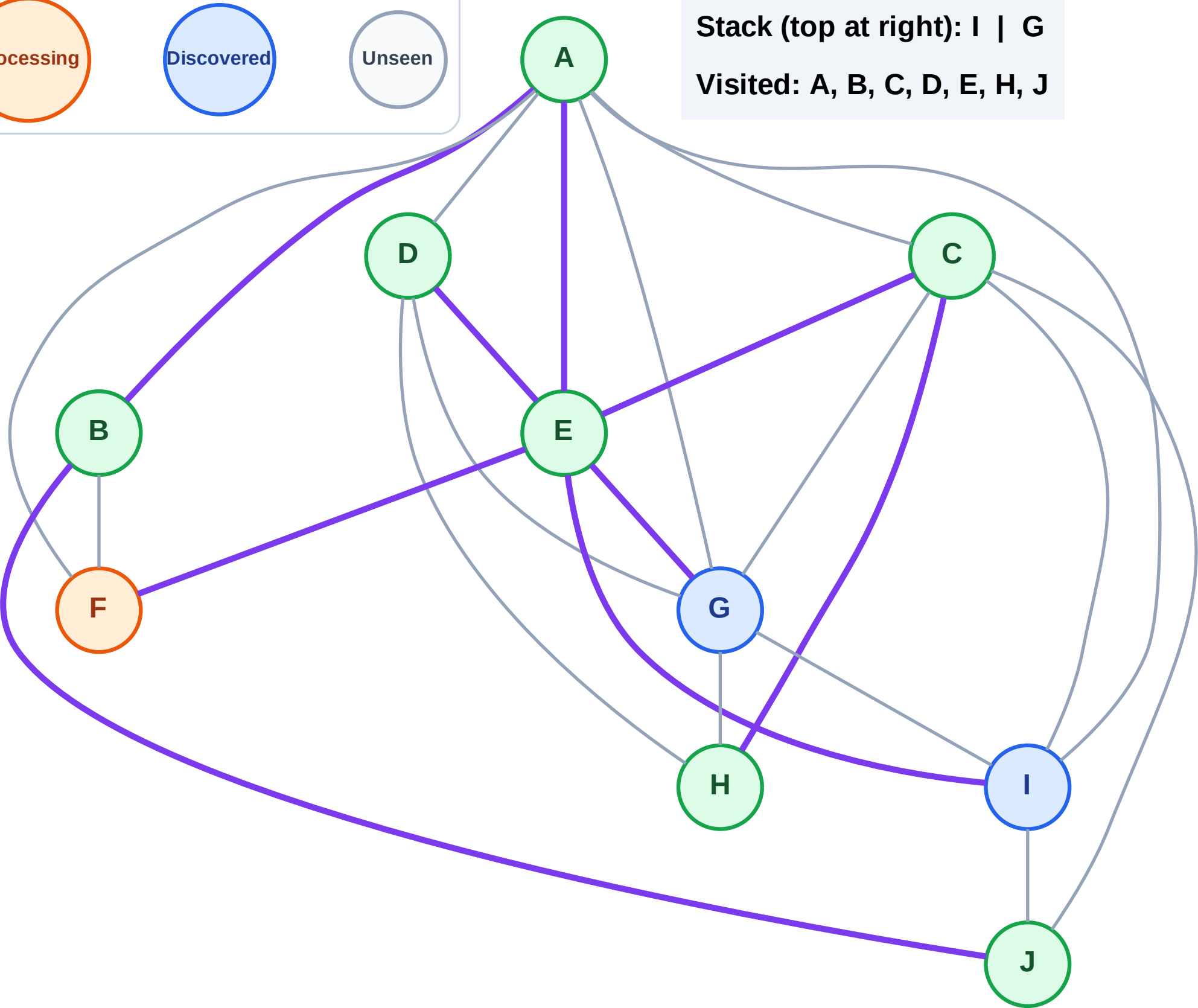
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G
Visited: A, B, C, D, E, H, J



DFS Traversal

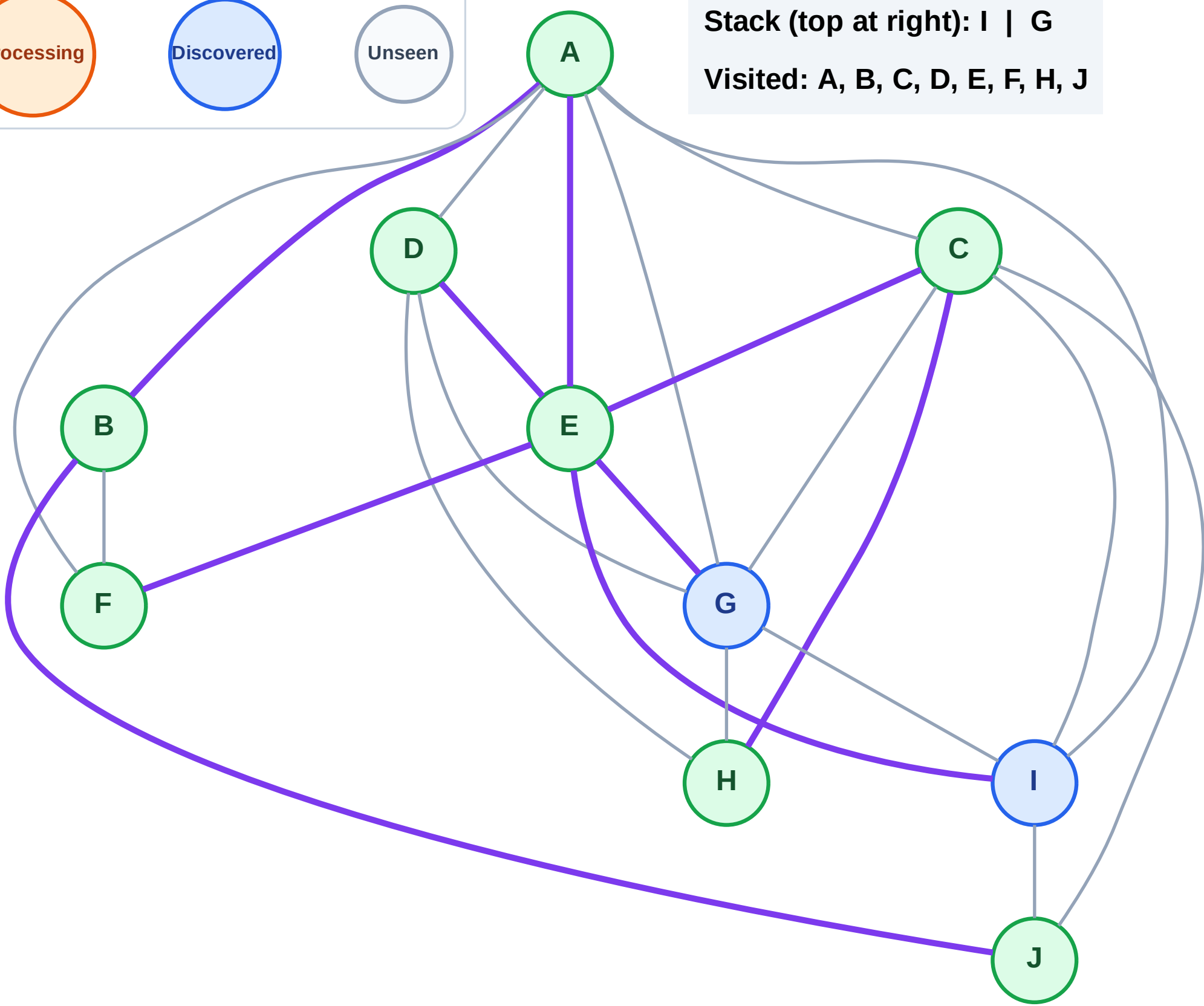
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

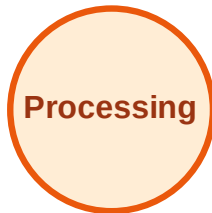
Visited: A, B, C, D, E, F, H, J



DFS Traversal

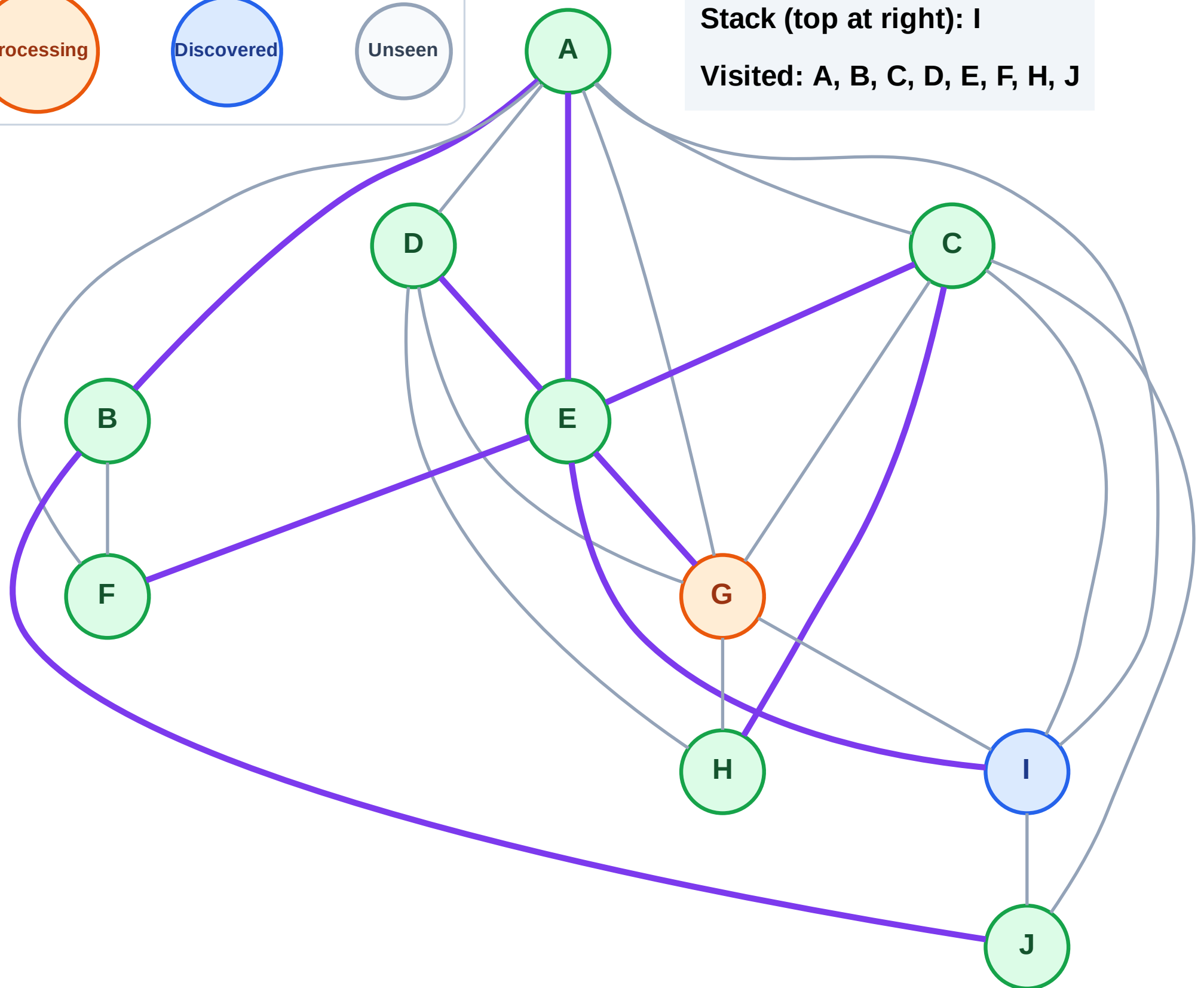
Step 18: Process node G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 19: No new neighbors from G

Legend

Complete

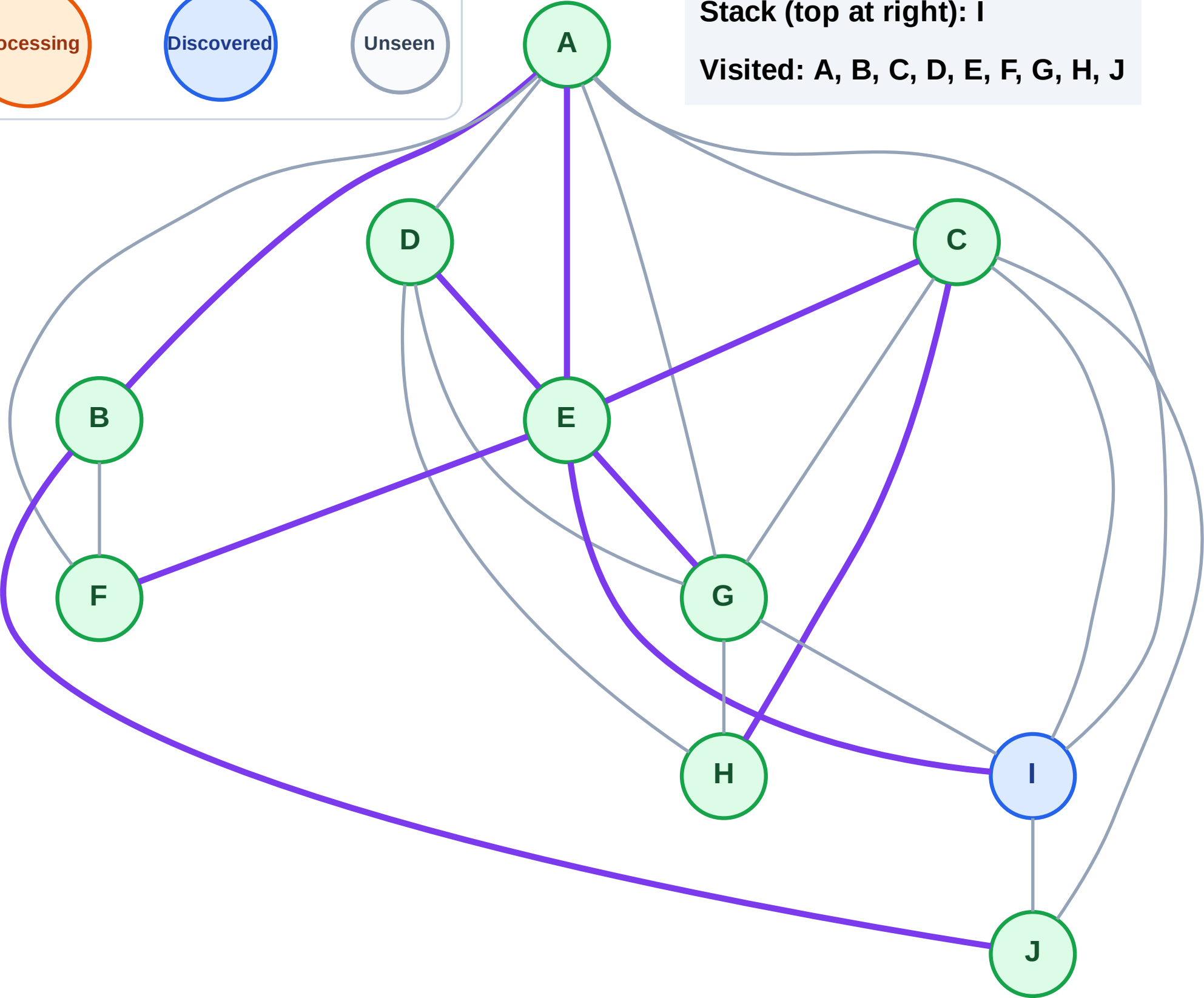
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

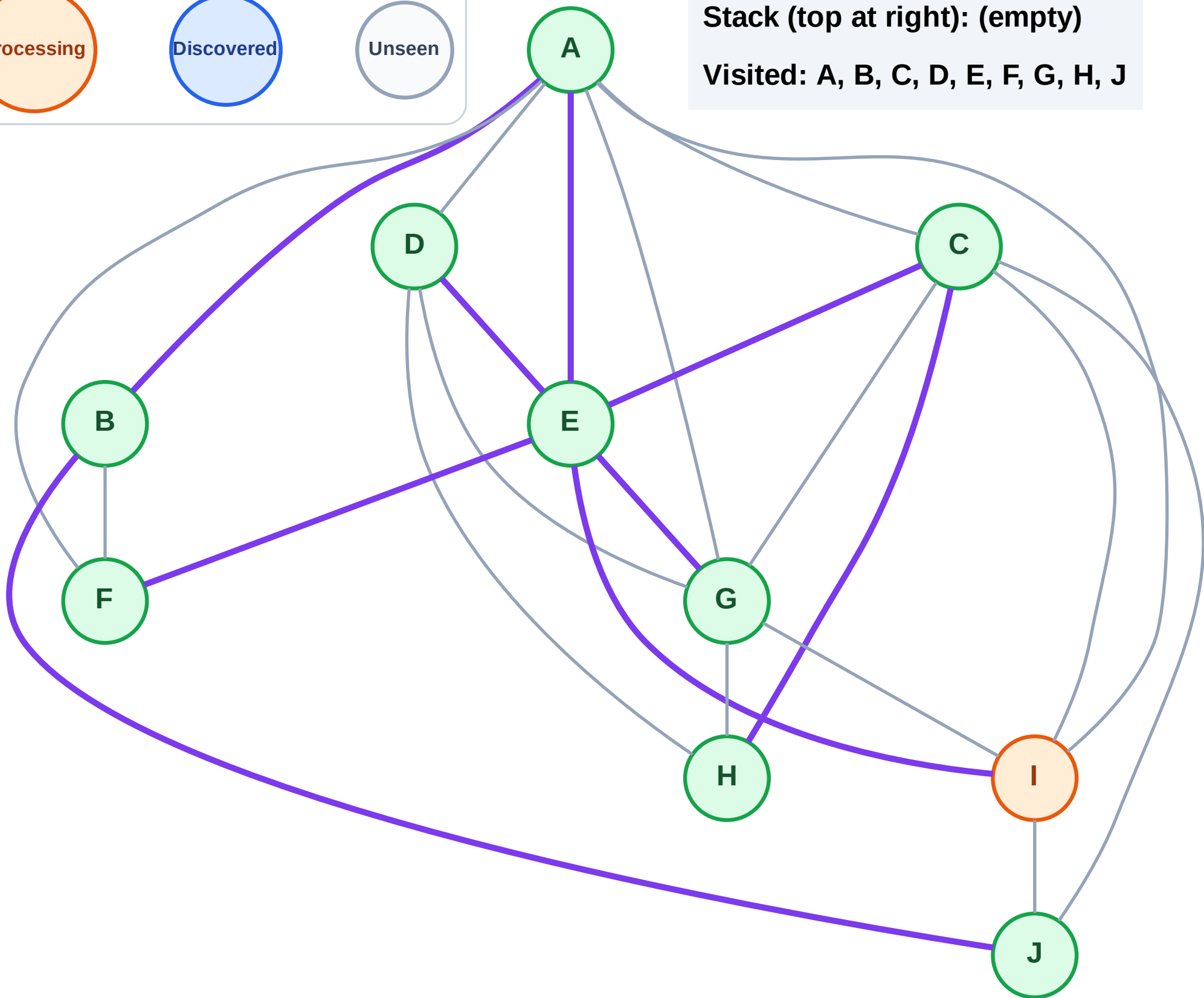
Step 20: Process node I

Legend



Stack (top at right): (empty)

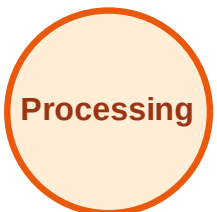
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

